

Cloud Compute Essentials

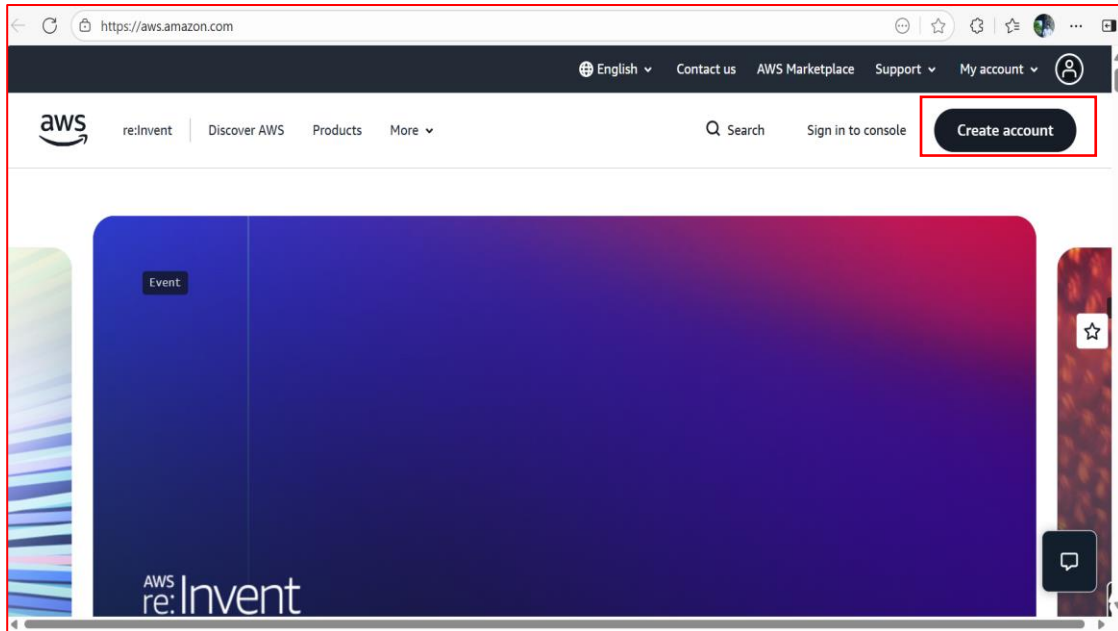
DevOps

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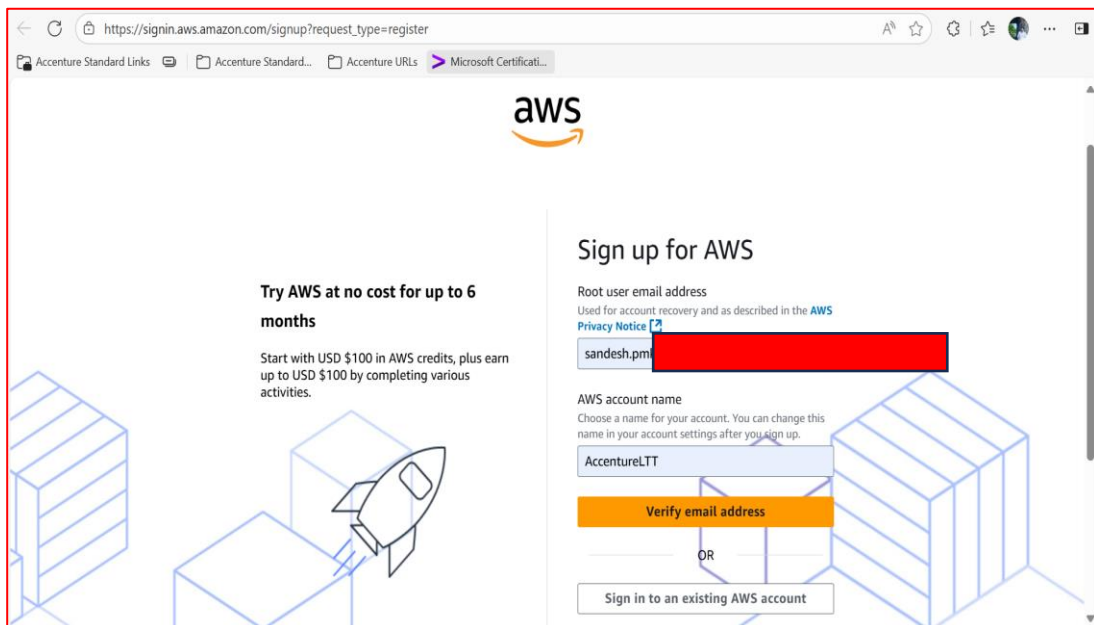
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1. Creating AWS account

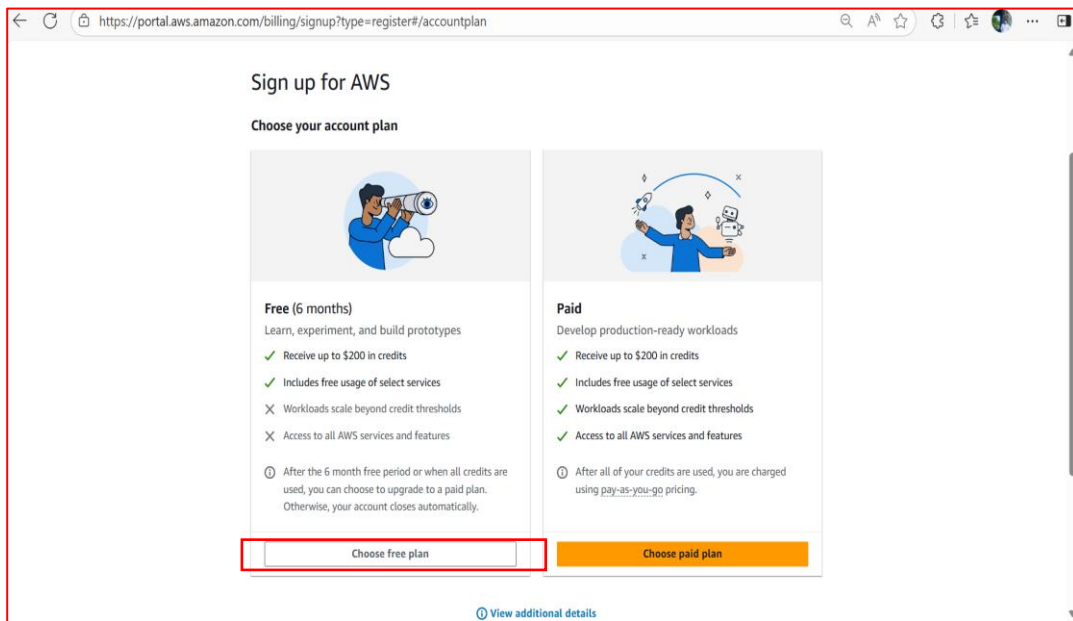
- 1 Sign up for AWS Account Using the Link below
[Cloud Computing Services - Amazon Web Services \(AWS\)](#)
Click on Create Account



- 2 Enter Email ID and AWS Account Name

A screenshot of the AWS sign-up page (https://signin.aws.amazon.com/signup?request_type=register). The page has a white background with the AWS logo at the top. On the left, there's a promotional text: 'Try AWS at no cost for up to 6 months' and 'Start with USD \$100 in AWS credits, plus earn up to USD \$100 by completing various activities.' Below this is a graphic of a rocket launching from a stack of blocks. On the right, the 'Sign up for AWS' section contains two input fields: 'Root user email address' (filled with 'sandesh.pmi') and 'AWS account name' (filled with 'AccentureLTT'). A 'Verify email address' button is below the email field. There's also an 'OR' section with a 'Sign in to an existing AWS account' link. A red box highlights the email input field.

3 Choose free plan



Sign up for AWS

Choose your account plan



Free (6 months)
Learn, experiment, and build prototypes

- ✓ Receive up to \$200 in credits
- ✓ Includes free usage of select services
- ✗ Workloads scale beyond credit thresholds
- ✗ Access to all AWS services and features

ⓘ After the 6 month free period or when all credits are used, you can choose to upgrade to a paid plan. Otherwise, your account closes automatically.

Choose free plan



Paid
Develop production-ready workloads

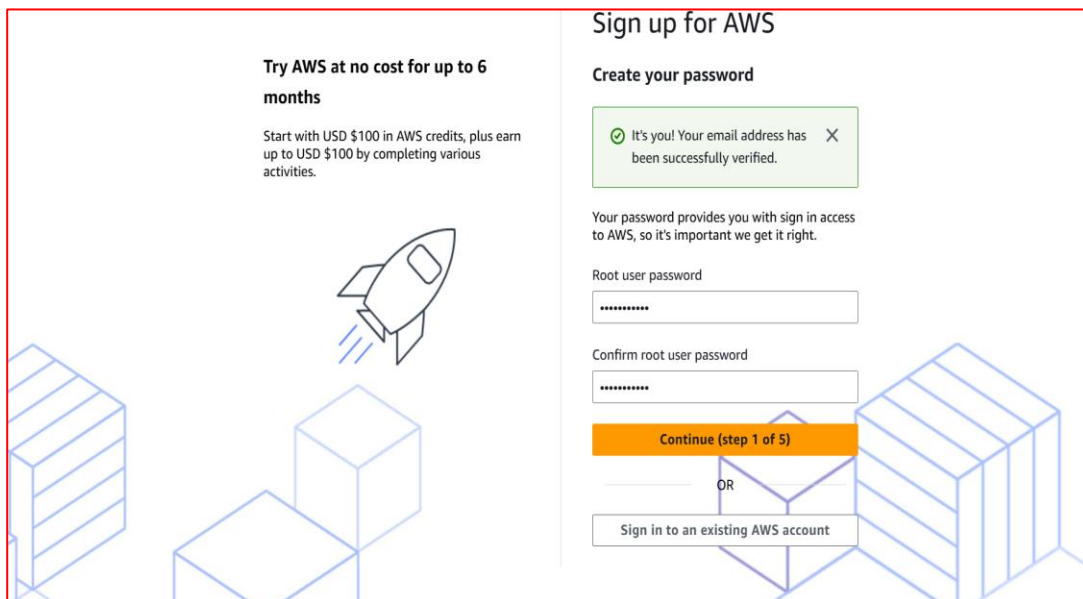
- ✓ Receive up to \$200 in credits
- ✓ Includes free usage of select services
- ✓ Workloads scale beyond credit thresholds
- ✓ Access to all AWS services and features

ⓘ After all of your credits are used, you are charged using pay-as-you-go pricing.

Choose paid plan

[View additional details](#)

4 Fill the details as it involves 5 steps for Sign up for AWS Step 1: Enter Password



Try AWS at no cost for up to 6 months

Start with USD \$100 in AWS credits, plus earn up to USD \$100 by completing various activities.

Sign up for AWS

Create your password

✓ It's you! Your email address has been successfully verified. ✕

Your password provides you with sign in access to AWS, so it's important we get it right.

Root user password

Confirm root user password

Continue (step 1 of 5)

OR

Sign in to an existing AWS account

Earn additional AWS credits

Complete various activities to earn up to an additional USD \$100 in credits, such as creating your first AWS budget to monitor cloud costs.

Contact Information

How do you plan to use AWS?

☐ Business - for your work, school, or organization

☒ Personal - for your own projects

Who should we contact about this account?

Full Name

Sandeep K.H

Country Code

+91

Phone Number

8495976330

Country or Region

India

Address line 1

Bangalore

Address line 2

Karnataka

City

Bangalore

State, Province, or Region

Karnataka

Postal Code

562166

Nearby AWS Region selection - optional

Enabling nearby start regions can provide benefits including improved performance. Uncheck the Region to prevent its usage. Learn more

☒ Enable Asia Pacific (Hyderabad) Region

Customers with an Indian contact address are served by Amazon Web Services India Private Limited, the local seller for AWS services in India.

☒ I have read and agree to the terms of the [AWS Customer Agreement](#)

Agree and Continue (step 2 of 6)

[illegible]

Step 4: Fill the details for Identity, you can Upload EPIC Copy of

1. Pan Card
2. Driving License
3. Voter ID

Upload Respective proof of Front Side and Back Side

Sign up for AWS

Confirm your identity

Primary purpose of account registration

Choose one that best applies to you. If your account is used to run a business, select the one that applies to your business.

Personal use

Ownership type

Choose your ownership relation to the account. Based on your selection, you may be required to complete additional customer verification.

Individual

India document type

To verify your identity, the name on the document must match the name that you chose.

Driving License

Date of birth

To use this document type, you must be at least 18 years old.

27/09/1992

Driving license number

Enter driving license number.

The driving license number is 12-18 alphanumeric characters without spaces or hyphens.

Name

Choose the name that you want to use for identity verification.

Sandeep K H

Upload front of Driving License

Choose file

Upload back of Driving License

Choose file

I consent to allowing AWS to use and send the information above to a third-party service for identity verification purposes.

Continue (step 4 of 5)

Step 4: Confirm your Identity by providing your number and Verification Code Sent to Register number

Sign up for AWS

Confirm your identity

Before you can use your AWS account, you must verify your phone number. When you continue, the AWS automated system will contact you with a verification code.

How should we send you the verification code?

☒ Text message (SMS)

☐ Voice call

Country or region code

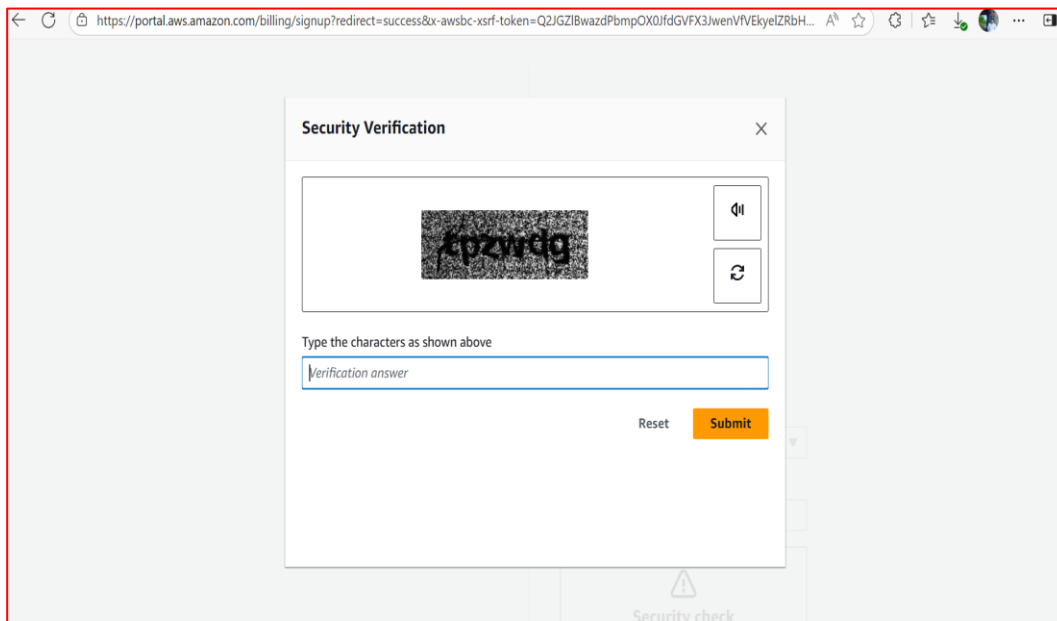
India (+91)

Mobile phone number

84959763

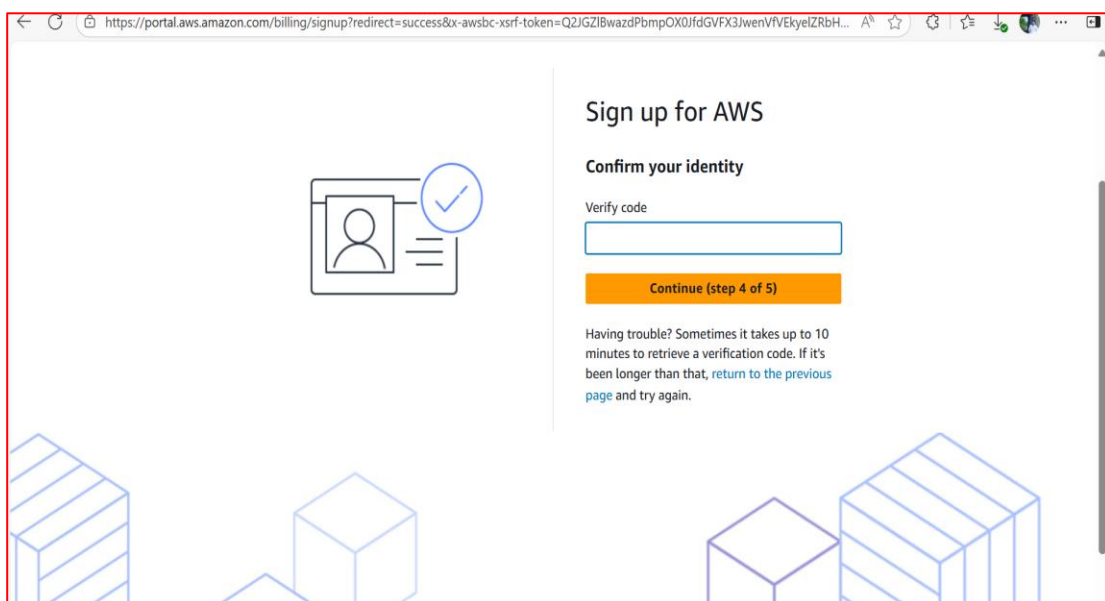
Send SMS (step 4 of 5)

5 Enter Verification



The screenshot shows a web browser window with the URL <https://portal.aws.amazon.com/billing/signup?redirect=success&awsbc-xsrf-token=Q2JGZlBwazdPbmpOX0JfdGVFX3JwenVlVEkyelZRbH...>. A modal dialog box titled "Security Verification" is centered on the screen. Inside the dialog, there is a CAPTCHA image showing the word "bpmzndg" in a noisy, pixelated font. To the right of the image are two buttons: a speaker icon for audio and a circular arrow for refresh. Below the image, the text "Type the characters as shown above" is followed by a text input field containing the placeholder text "Verification answer". At the bottom of the dialog are "Reset" and "Submit" buttons. The background of the browser window is dimmed, showing a "Security check" warning icon.

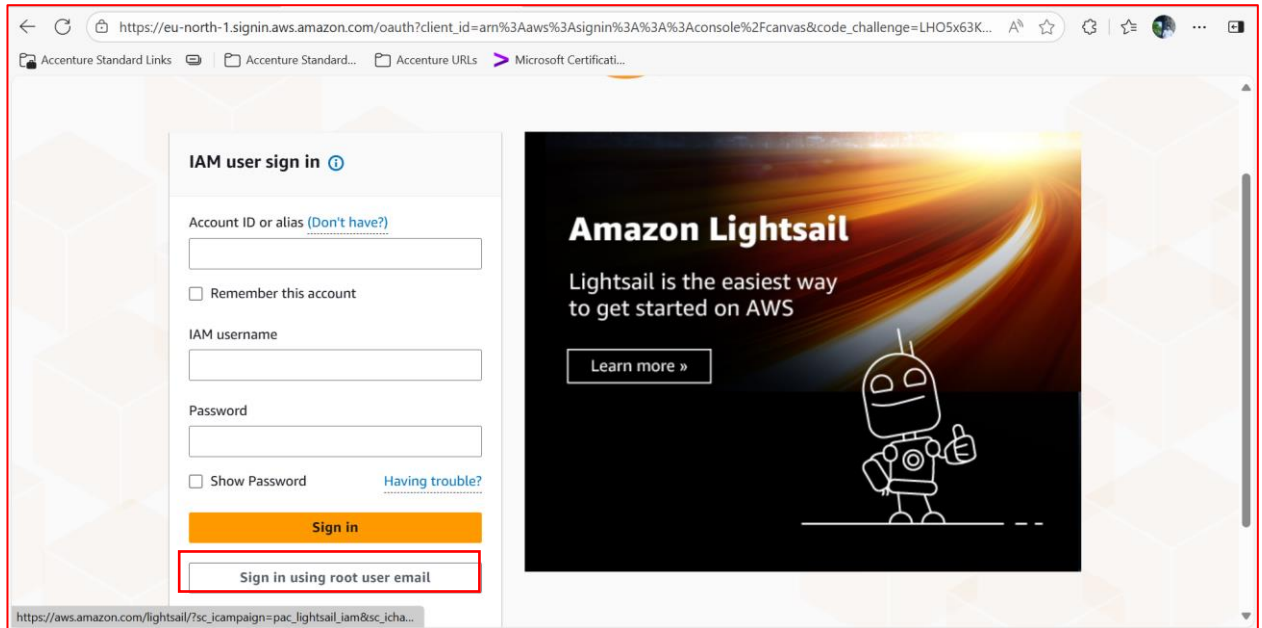
Enter the text or voice verification code that you have received



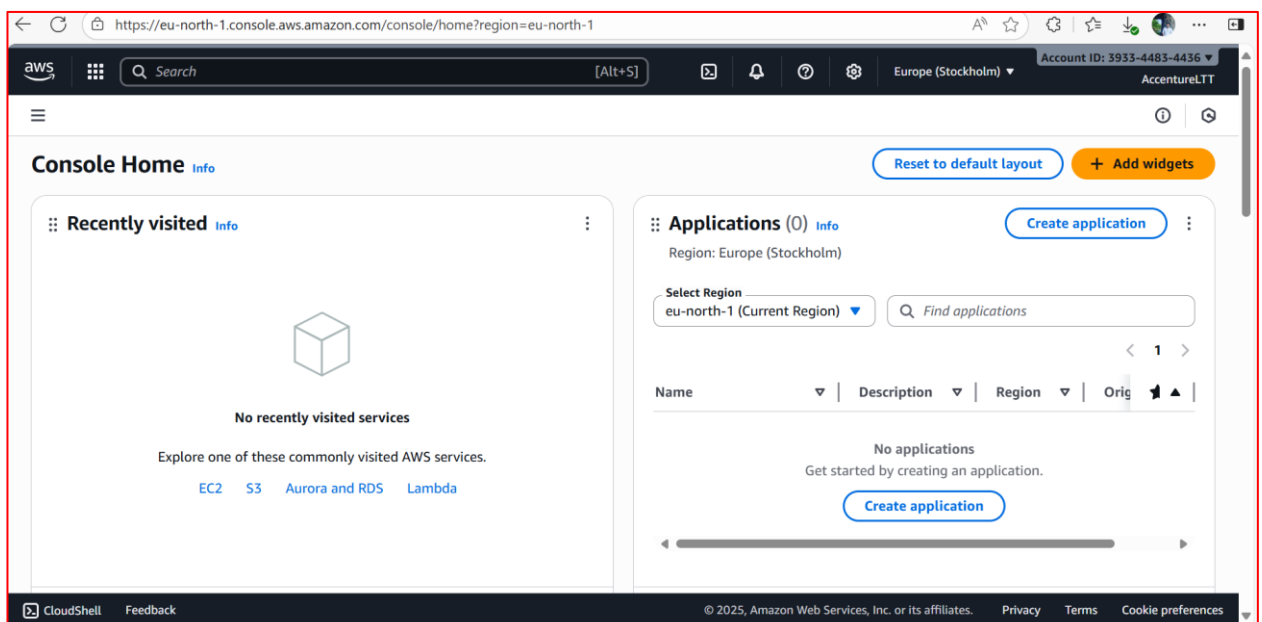
The screenshot shows the "Sign up for AWS" page. On the left, there is an illustration of a person's profile with a checkmark, indicating successful verification. On the right, the heading "Sign up for AWS" is followed by the sub-heading "Confirm your identity". Below this, there is a "Verify code" label and a text input field. An orange button labeled "Continue (step 4 of 5)" is positioned below the input field. A note at the bottom states: "Having trouble? Sometimes it takes up to 10 minutes to retrieve a verification code. If it's been longer than that, [return to the previous page](#) and try again." The page features a decorative footer with blue wireframe cubes.

2. Creating EC2 Instance on AWS

- 1 **Sign in to AWS Management Console**
<https://aws.amazon.com/console/>
 1. Click **"Sign in to the Console."**
 2. Choose:
 - **Root user**-use your email and password

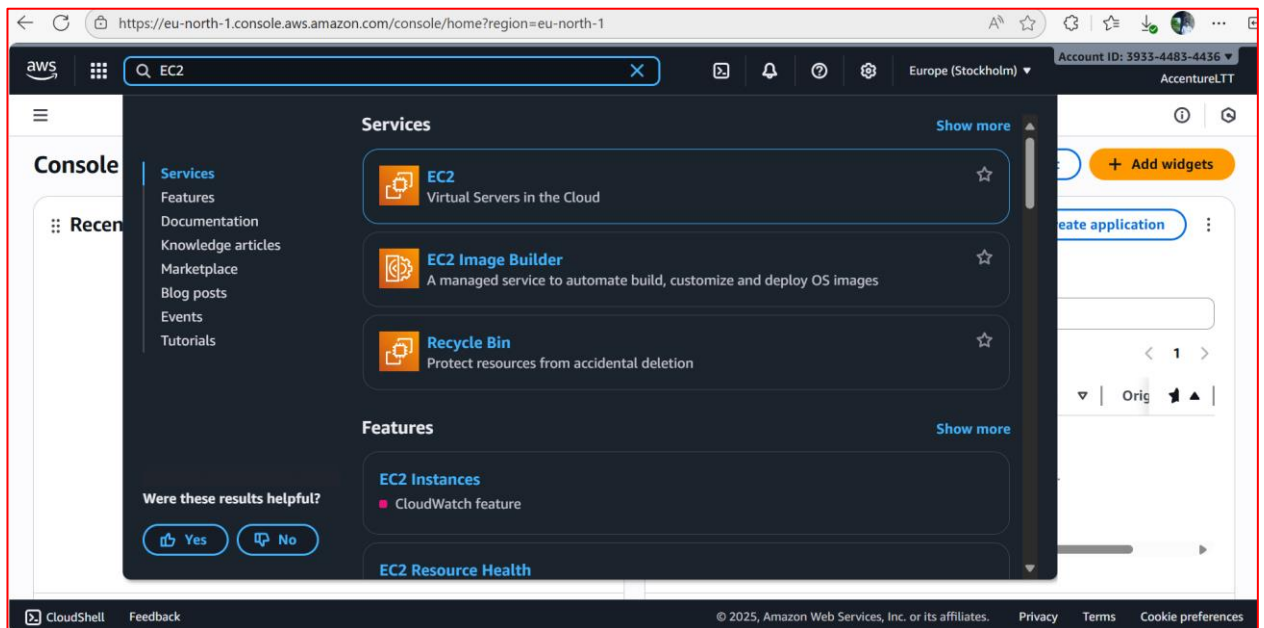


- 2 **AWS Management Console Dashboard.**



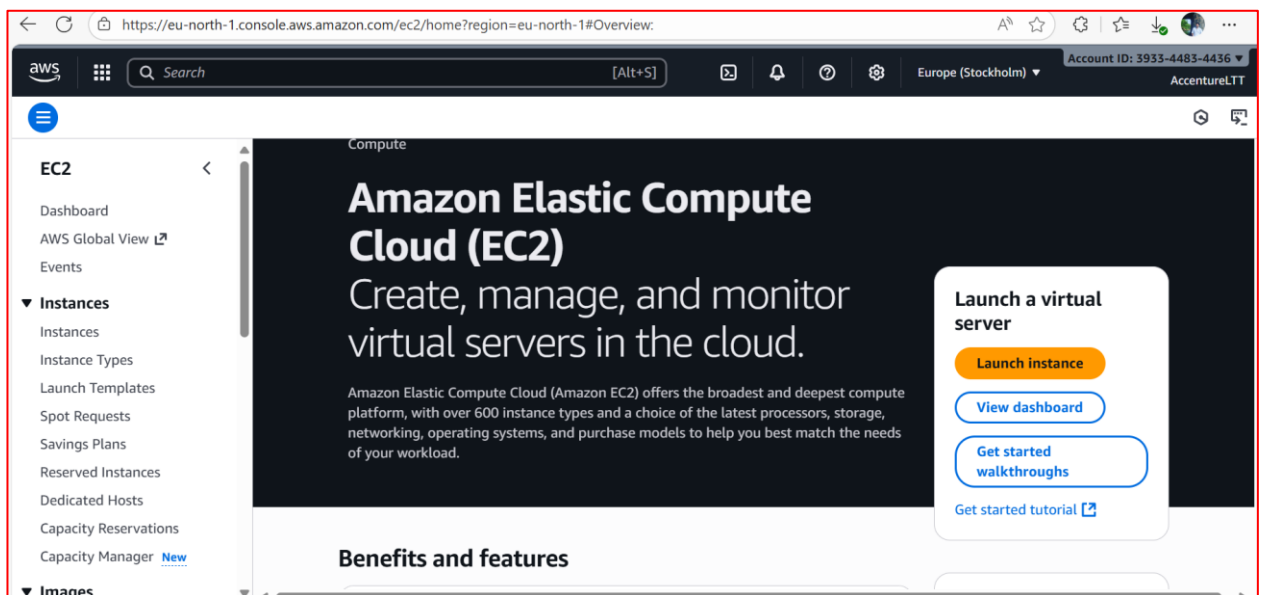
3 Open EC2 Dashboard

In the **search bar** at the top, type “EC2”.
Click **EC2** under “Services.”

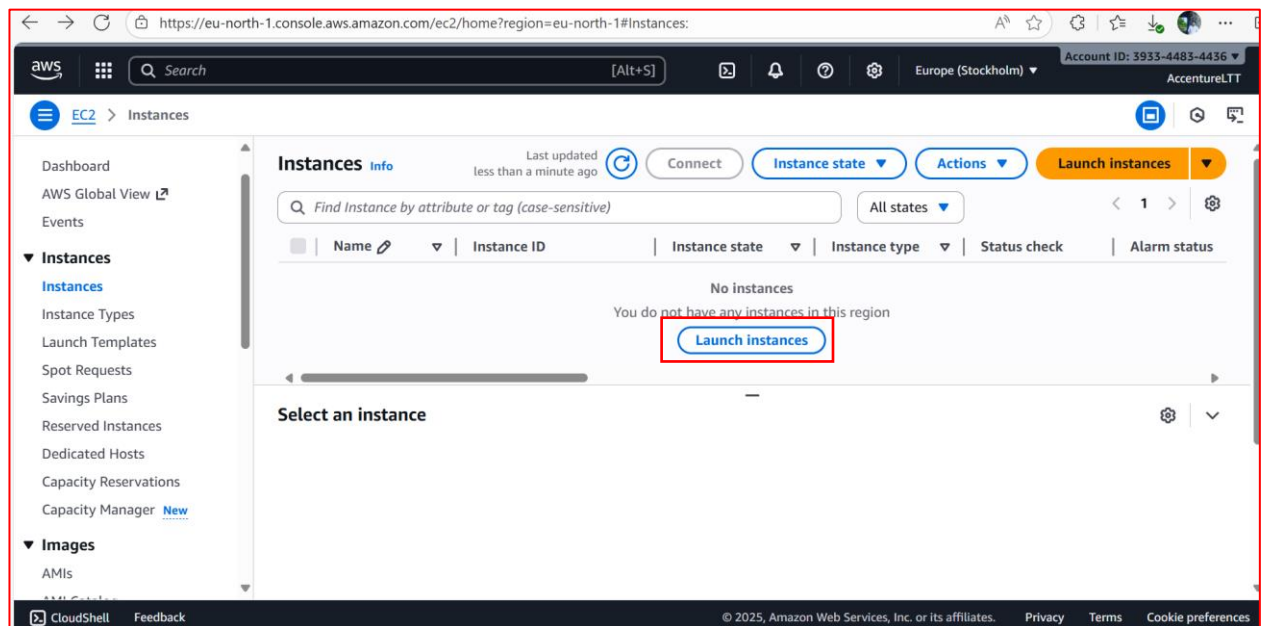


You'll see the **EC2 Dashboard**, showing:

- Instances
- Volumes
- Security Groups
- Key Pairs, etc.
-



4 **Launch a New EC2 Instance**
Click “Launch instance.”



- 5 Fill the details:
Name and tags → e.g., MyFirstEC2
Application and OS Image (AMI) → Ubuntu. (Free tier eligible)
Instance type → Choose t3.micro (free tier eligible)

Launch an instance

Name and tags

Name: MYFirstEC2

Application and OS Images (Amazon Machine Image)

Quick Start

Amazon Linux, macOS, **Ubuntu**, Windows, Red Hat, SUSE Linux, Debian

Amazon Machine Image (AMI)

Ubuntu Server 24.04 LTS (HVM), SSD Volume Type

Description: Ubuntu Server 24.04 LTS (HVM), SSD Volume Type. Support available from Canonical (http://www.ubuntu.com/cloud/services).

Architecture: 64-bit (x86), AMI ID: ami-0a716d3f3b16d290c, Publish Date: 2025-08-21, Username: ubuntu

Instance type

t3.micro

Summary

Number of instances: 1

Software Image (AMI): Canonical, Ubuntu, 24.04, amd64...

Virtual server type (instance type): t3.micro

Firewall (security group): New security group

Storage (volumes): 1 volume(s) - 8 GiB

Launch instance

6 As we are using Ubuntu use Key pair type as RSA and File format .pem

Launch an instance

Key pair (login)

Key pair name - required: MyFirstEC2-KeyPair

Network settings

Network: vpc-078b52b4769684743

Subnet: No preference (Default subnet in any availability zone)

Auto-assign public IP: Enable

Firewall (security groups): Create security group

Create key pair

Key pair name: MyFirstEC2-KeyPair

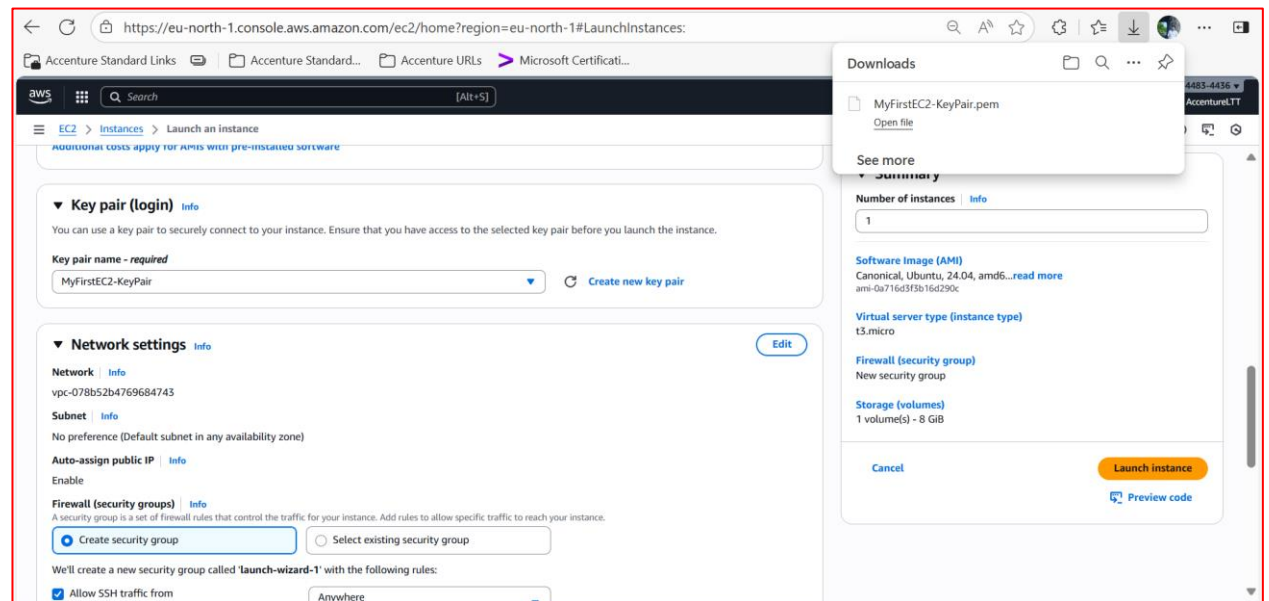
Key pair type: **RSA**

Private key file format: **.pem**

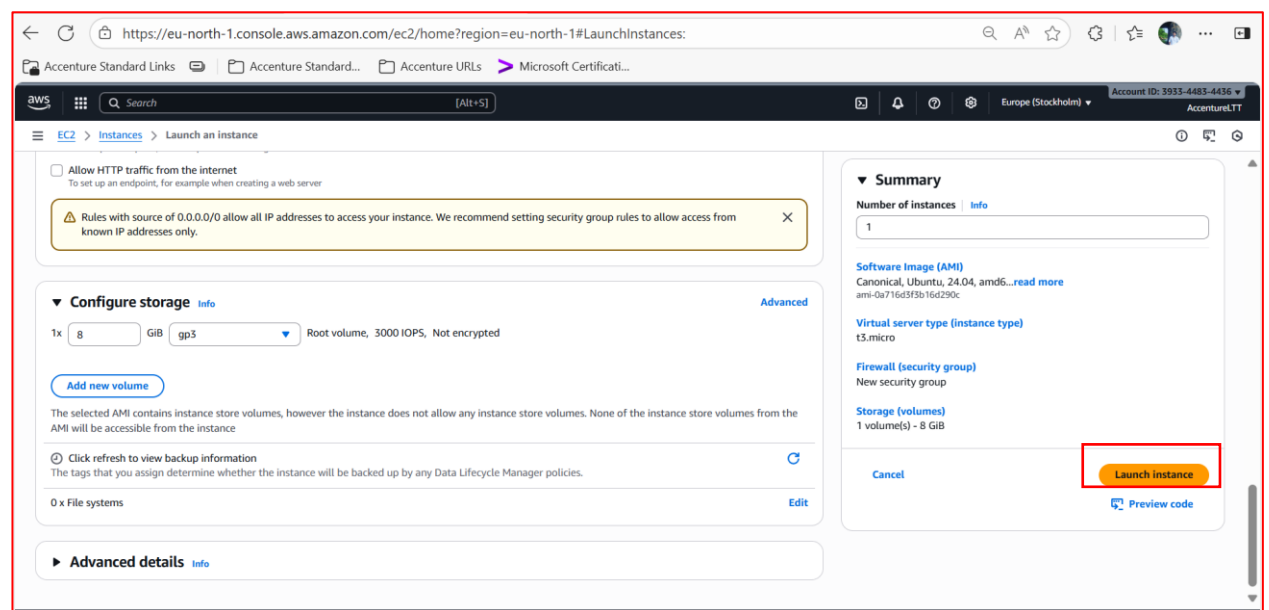
When prompted, store the private key in a secure and accessible location on your computer. You will need it later to connect to your instance.

Launch instance

.pem File will be downloaded once we key pair is created



7 Launch the instance



After Success

The screenshot shows the AWS Management Console interface for the 'Launch an instance' page. A green success banner at the top states: 'Success Successfully initiated launch of instance (i-09c04653c7d20d2be)'. Below this is a 'Launch log' section. The 'Next Steps' section provides a search bar and several actionable cards:

- Create billing usage alerts:** To manage costs and avoid surprise bills, set up email notifications for billing usage thresholds. [Create billing alerts](#)
- Connect to your instance:** Once your instance is running, log into it from your local computer. [Connect to instance](#) [Learn more](#)
- Connect an RDS database:** Configure the connection between an EC2 instance and a database to allow traffic flow between them. [Connect an RDS database](#) [Create a new RDS database](#) [Learn more](#)
- Create EBS snapshot policy:** Create a policy that automates the creation, retention, and deletion of EBS snapshots. [Create EBS snapshot policy](#)
- Manage detailed monitoring:** Enable or disable detailed monitoring for the instance.
- Create Load Balancer:** Create an application, network gateway or classic Elastic Load Balancing.
- Create AWS budget:** AWS Budgets allows you to create budgets, forecast, and set alerts.
- Manage CloudWatch alarms:** Create or update Amazon CloudWatch alarms for the instance.

Click on Instance, can observe the instance running

The screenshot shows the 'Instance summary' page for instance **i-09c04653c7d20d2be (MYFirstEC2)**. The instance state is **Running**, which is highlighted with a red box. The summary includes the following details:

- Instance ID:** i-09c04653c7d20d2be
- Public IPv4 address:** 13.62.127.78 | [open address](#)
- Private IPv4 addresses:** 172.31.27.186
- Public DNS:** ec2-13-62-127-78.eu-north-1.compute.amazonaws.com | [open address](#)
- Instance state:** Running
- Private IP DNS name (IPv4 only):** ip-172-31-27-186.eu-north-1.compute.internal
- Instance type:** t3.micro
- VPC ID:** vpc-078b52b4769684743
- Subnet ID:** subnet-0adbe56fba7dd7240
- Instance ARN:** arn:aws:ec2:eu-north-1:393344834436:instance/i-09c04653c7d20d2be
- Hostname type:** IP name: ip-172-31-27-186.eu-north-1.compute.internal
- Answer private resource DNS name:** IPv4 (A)
- Auto-assigned IP address:** 13.62.127.78 [Public IP]
- IAM Role:** -
- IMDSv2:** Required
- Elastic IP addresses:** -
- AWS Compute Optimizer finding:** Opt-in to AWS Compute Optimizer for recommendations. | [Learn more](#)
- Auto Scaling Group name:** -
- Managed:** false

By Default, it should have all Inbound rules as shown below

Successfully initiated stopping of i-09c04653c7d20d2be

Instances (1/1) info

Find Instance by attribute or tag (case-sensitive) All states

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone
MYFirstEC2	i-09c04653c7d20d2be	Stopped	t3.micro	3/3 checks passed	View alarms +	eu-north-1a

i-09c04653c7d20d2be (MYFirstEC2)

Name	Security group rule ID	Port range	Protocol	Source	Source type
-	sgr-0486c0c0ed8ad9fdf	8080	TCP	0.0.0.0/0	la
-	sgr-0dfae261a3aa86e59	22	TCP	0.0.0.0/0	la
-	sgr-0d1d17d8b77f666bf	80	TCP	0.0.0.0/0	la

- 8 If it is not available, you can add Inbound rules,
Go to EC2 Dashboard→Instance→Click on Instance→Security Tab→Security Groups

Instances (1/1) info

Find Instance by attribute or tag (case-sensitive) All states

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone
MYFirstEC2	i-09c04653c7d20d2be	Stopped	t3.micro	-	View alarms +	eu-north-1a

i-09c04653c7d20d2be (MYFirstEC2)

IAM Role -

Owner ID 393344834436

Launch time Mon Nov 03 2025 10:56:49 GMT+0530 (India Standard Time)

Security groups sg-02a4f89d620b87d45 (launch-wizard-1)

Inbound rules

Scroll down

The screenshot shows the AWS Management Console for an EC2 instance. The 'Security' tab is selected, and the 'Security groups' section is highlighted with a red box. The 'Inbound rules' section is also visible, showing a table of rules.

Security details

- IAM Role:** -
- Owner ID:** 393344834436
- Launch time:** Mon Nov 03 2025 10:56:49 GMT+0530 (India Standard Time)

Security groups

- [sg-02a4f89d620b87d45 \(launch-wizard-1\)](#)

Inbound rules

Name	Security group rule ID	Port range	Protocol	Source
-	sgr-0486c0c0ed8ad9fdf	8080	TCP	0.0.0.0/0
-	sgr-0dfae261a3aa86e59	22	TCP	0.0.0.0/0

9 Once you click on Security Groups Launch wizard, you will find an option to Edit Inbound rules

The screenshot shows the AWS Management Console for a Security Group. The 'Inbound rules' tab is selected, and the 'Edit inbound rules' button is highlighted with a red box. The 'Inbound rules (3)' section is visible, showing a table of rules.

Details

- Security group name:** launch-wizard-1
- Security group ID:** sg-02a4f89d620b87d45
- Description:** launch-wizard-1 created 2025-10-31T04:52:06.526Z
- VPC ID:** vpc-078b52b4769684743
- Owner:** 393344834436
- Inbound rules count:** 3 Permission entries
- Outbound rules count:** 1 Permission entry

Inbound rules (3)

Name	Security group rule ID	IP version	Type	Protocol	Port range
-	sgr-0486c0c0ed8ad9fdf	IPv4	Custom TCP	TCP	8080
-	sgr-0dfae261a3aa86e59	IPv4	SSH	TCP	22
-	sgr-0d1d17d8b77f666bf	IPv4	HTTP	TCP	80

- 10 Click on Add rule to add respective rules as per requirements.
By default, you can add Custom TCP, SSH and HTTP

https://eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#ModifyInboundSecurityGroupRules:securityGroupId=sg-02a4f89d620b87d45

Account ID: 3933-4483-4436

EC2 > Security Groups > sg-02a4f89d620b87d45 - launch-wizard-1 > Edit inbound rules

Edit inbound rules

Inbound rules control the incoming traffic that's allowed to reach the instance.

Security group rule ID	Type	Protocol	Port range	Source	Description - optional	Info
sg-0486c0c0ed8ad9fdf	Custom TCP	TCP	8080	Cust...	0.0.0.0/0	Delete
sg-0dfae261a3aa86e59	SSH	TCP	22	Cust...	0.0.0.0/0	Delete
sg-0d1d17d8b77f666bf	HTTP	TCP	80	Cust...	0.0.0.0/0	Delete

[Add rule](#)

Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

- 11 You can connect your instance on CLI in AWS account by clicking on Connect

https://eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#Instances:

Account ID: 3933-4483-4436

EC2 > Instances

Instances (1/1)

Find Instance by attribute or tag (case-sensitive)

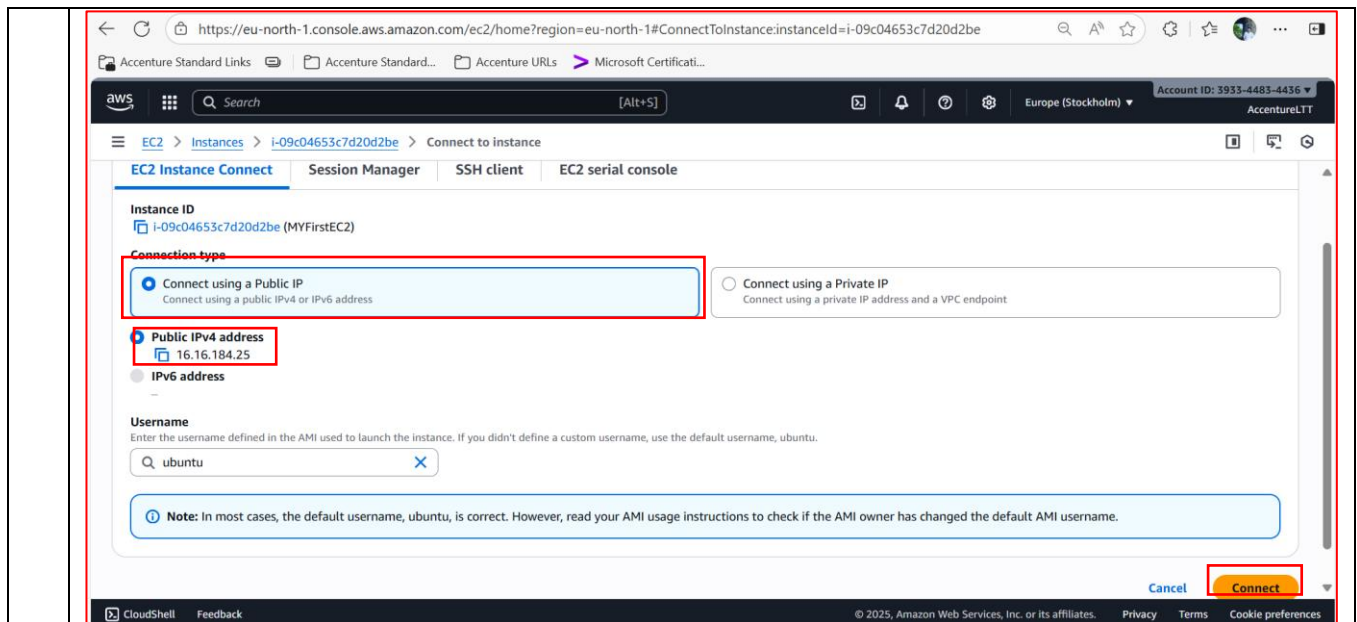
Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone
MYFirstEC2	i-09c04653c7d20d2be	Running	t3.micro	Initializing	View alarms +	eu-north-1a

i-09c04653c7d20d2be (MYFirstEC2)

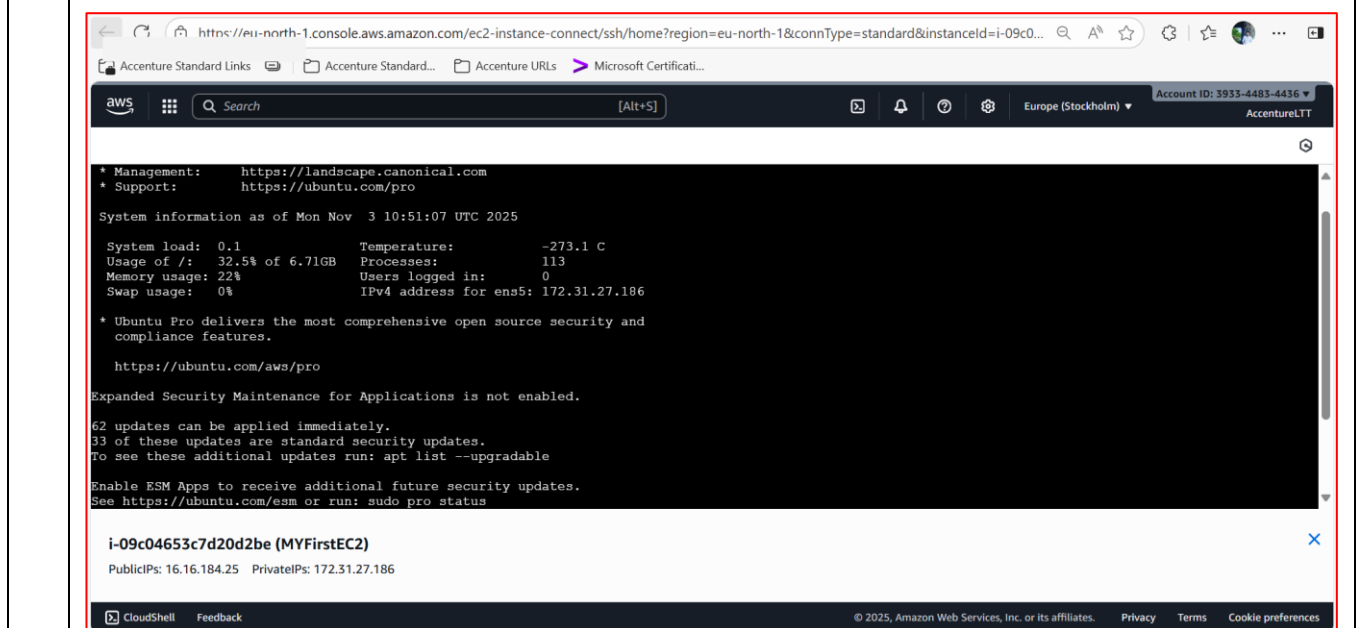
[Details](#) [Status and alarms](#) [Monitoring](#) [Security](#) [Networking](#) [Storage](#) [Tags](#)

Instance summary

Instance ID	Public IPv4 address	Private IPv4 addresses
i-09c04653c7d20d2be	16.16.184.25 open address	172.31.27.186
IPv6 address	Instance state	Public DNS



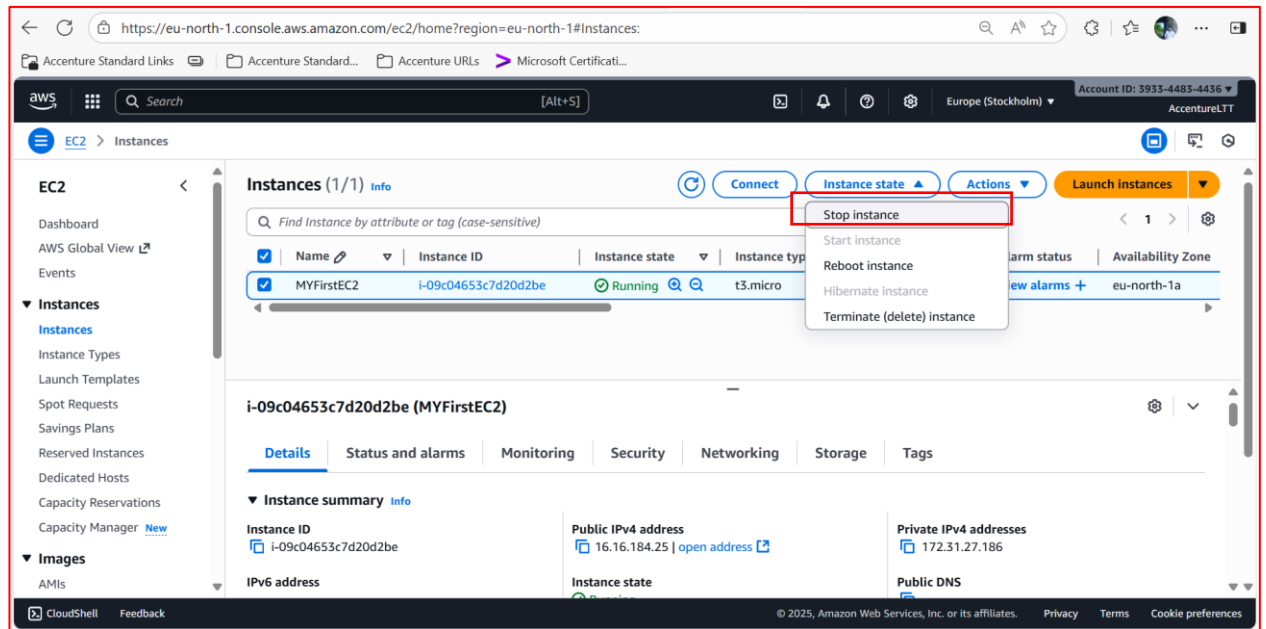
- 12 After successfully connecting your instance will be ready to use with CLI(Optional) using which, we can instance Jenkins, Kubernetes etc.



13 This is how we can create EC2 on AWS.

Note: - If you don't want any charges to be applied for your AWS account, make sure that you stop or terminate your instance when not in use

Click on Instance State → Stop Instance (In Future if you need) or Terminate Instance if not be used in Future

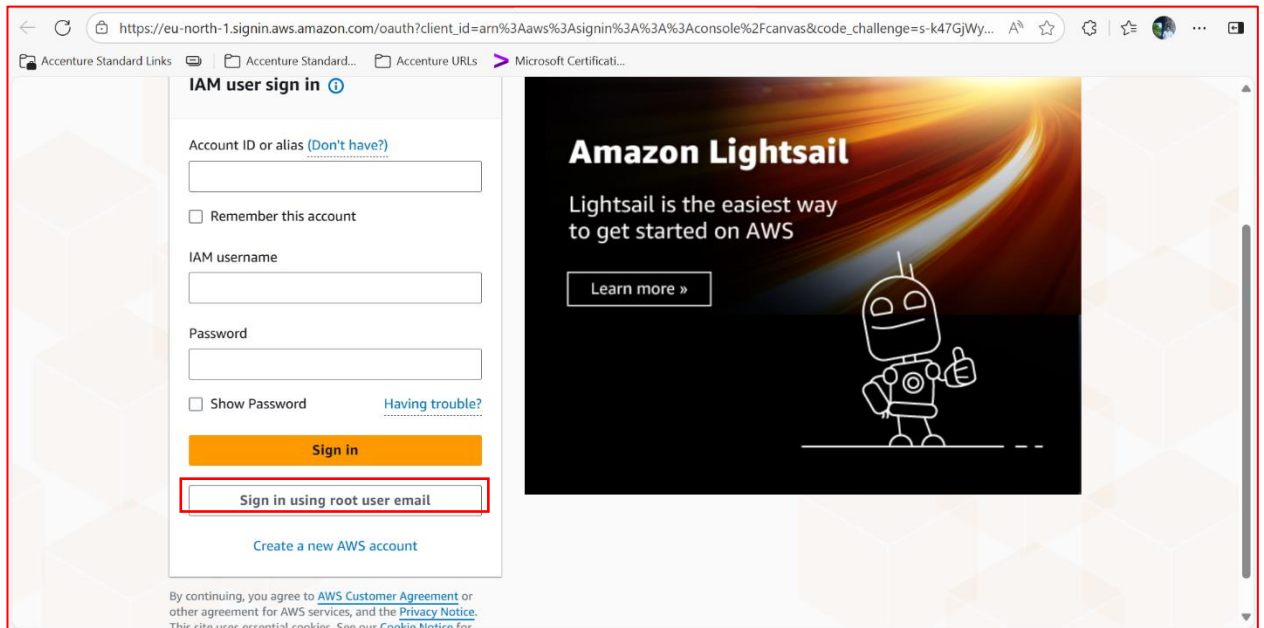


Thank You!

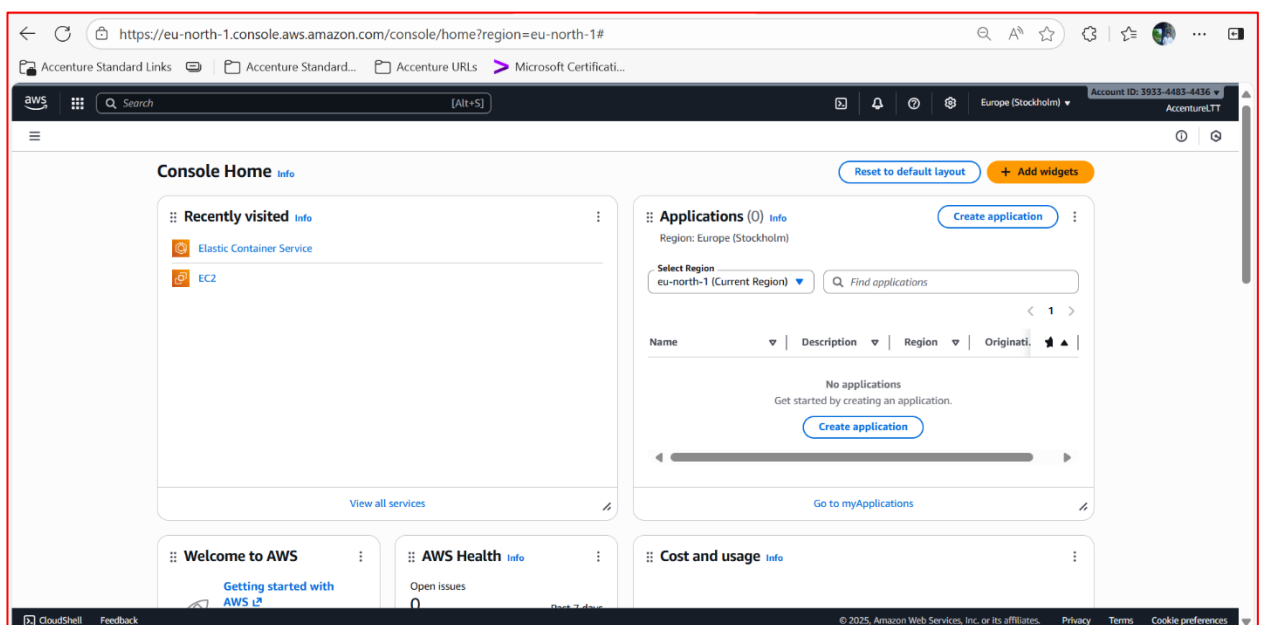
3. Create Amazon ECS (Elastic Container Service)

1 Use the link below to Sign-In

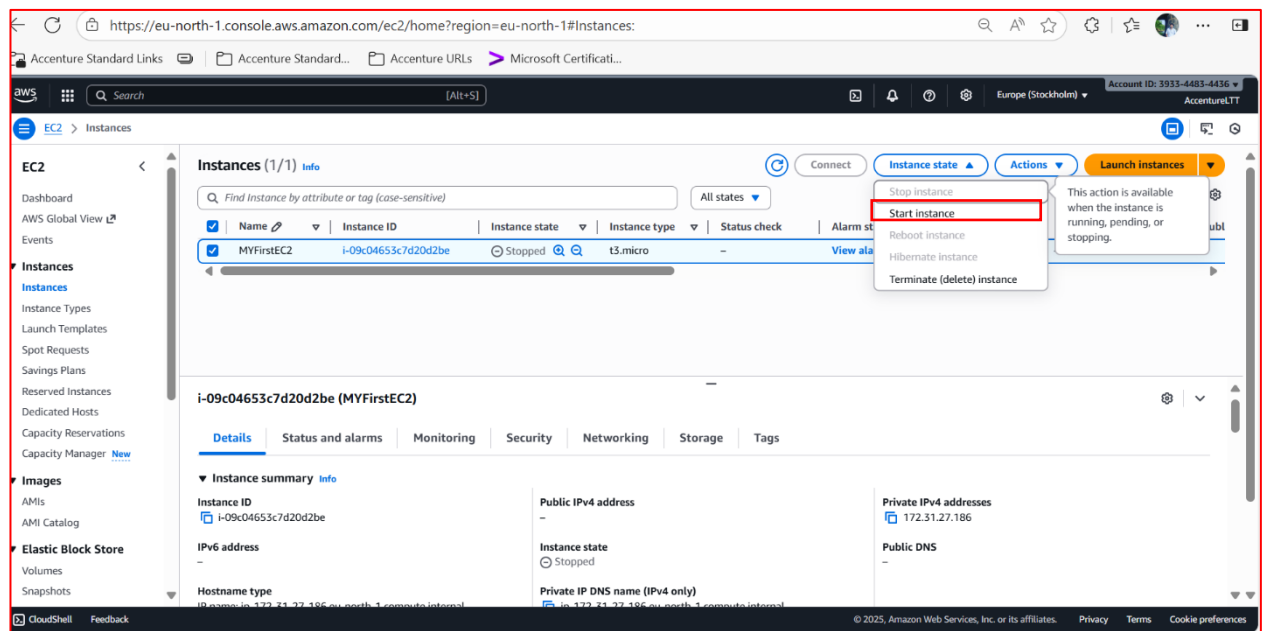
<https://console.aws.amazon.com/>



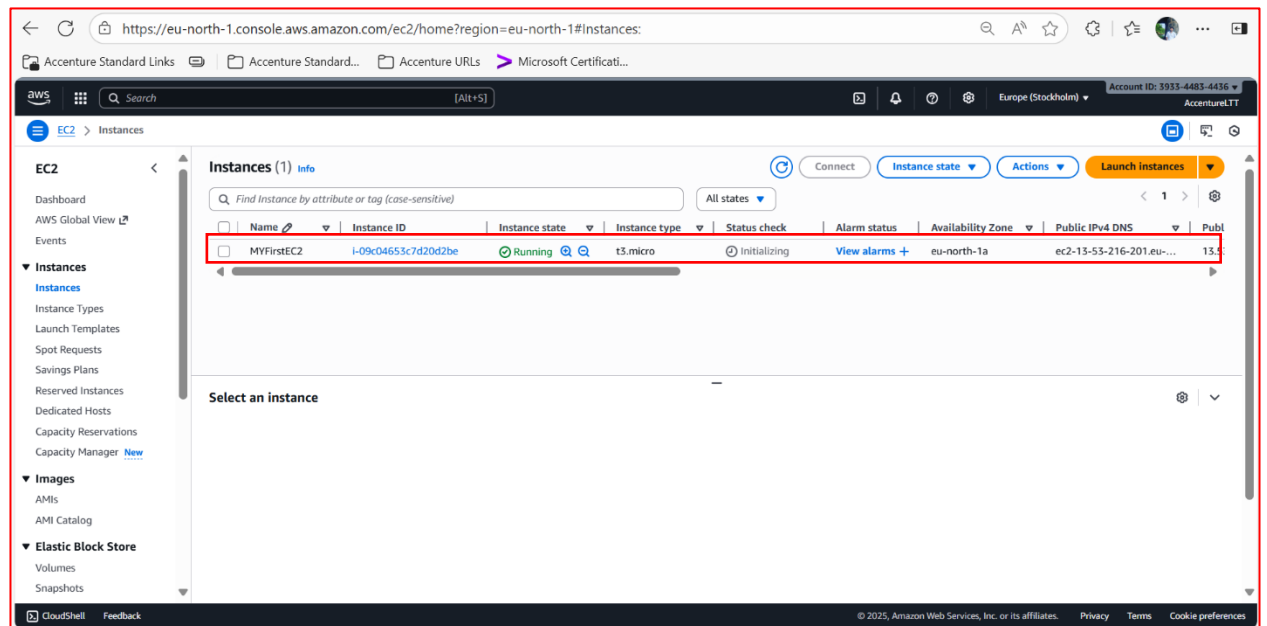
2 After Successful Sign In you will be launched to the AWS Dashboard



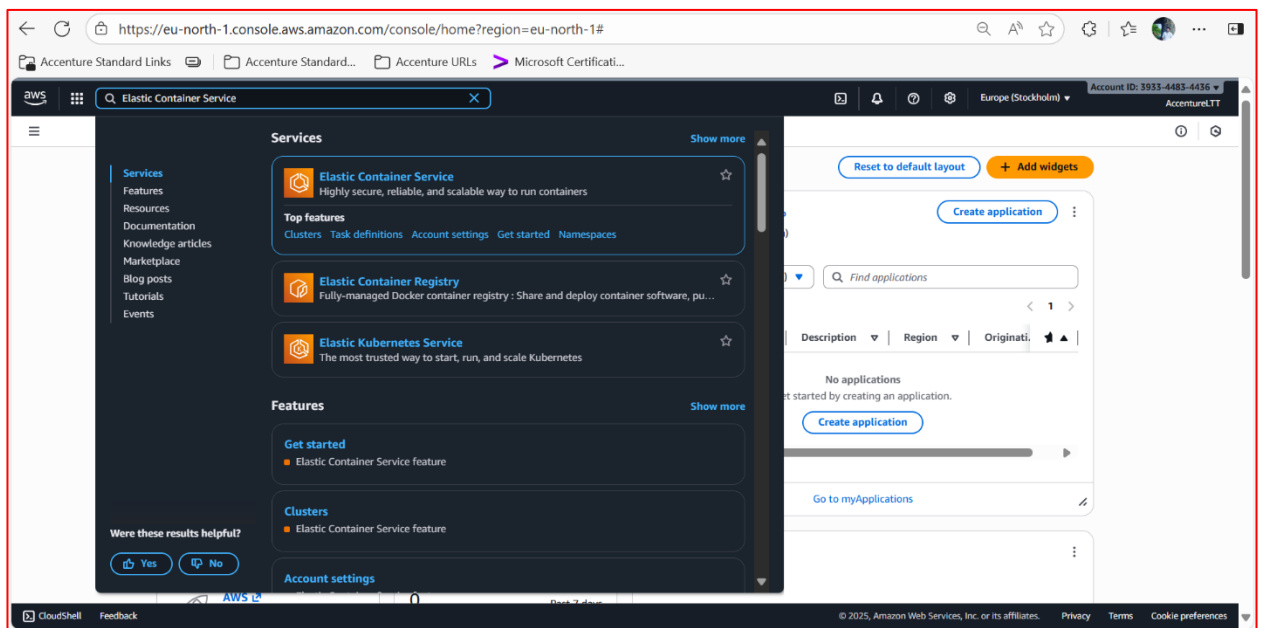
3 Before creating the ECS, launch the EC2 Instance created in previous steps



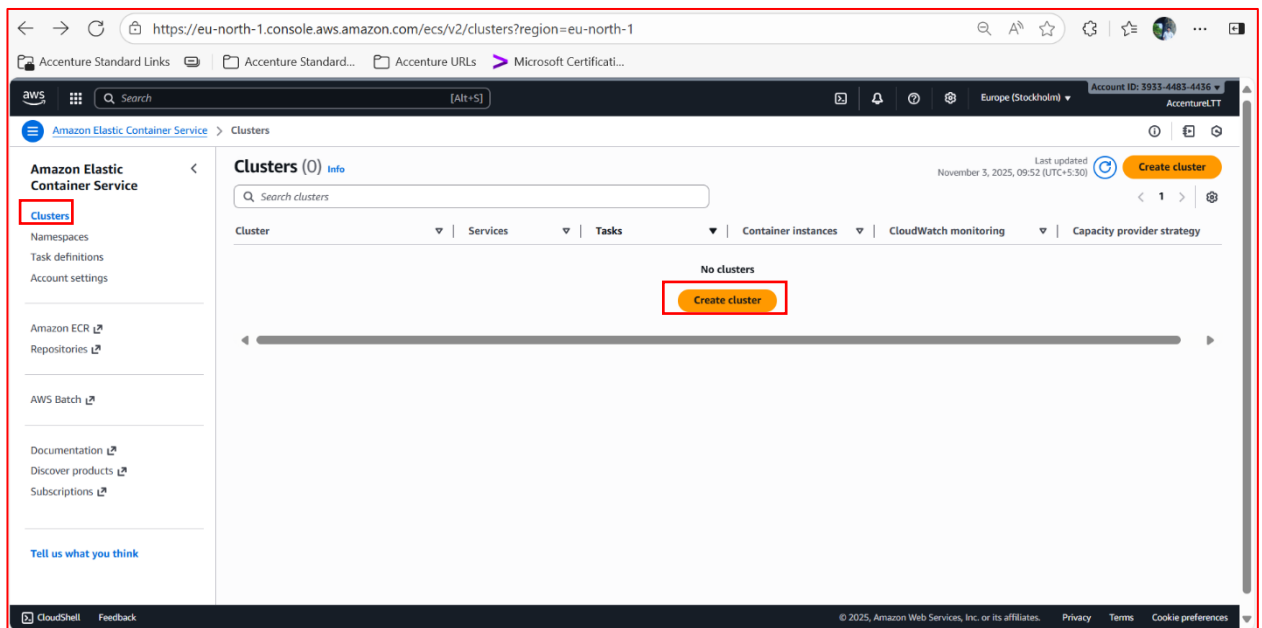
Make Sure that the Instance is running



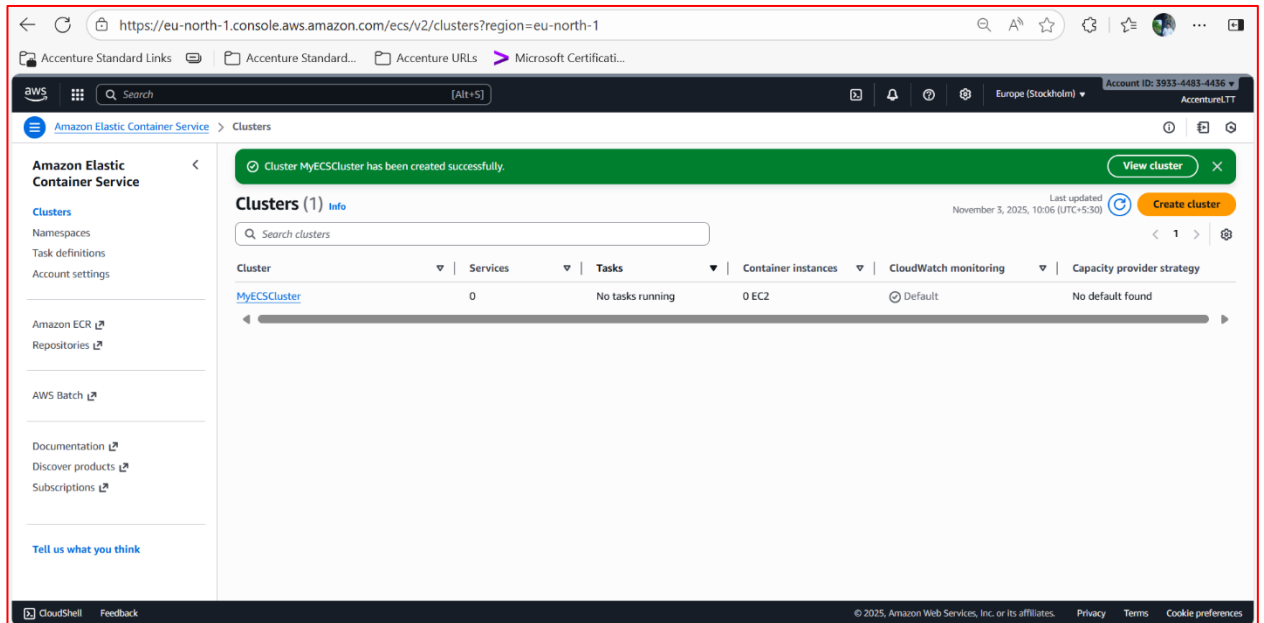
4 Search for Elastic Container Service and Click on the result



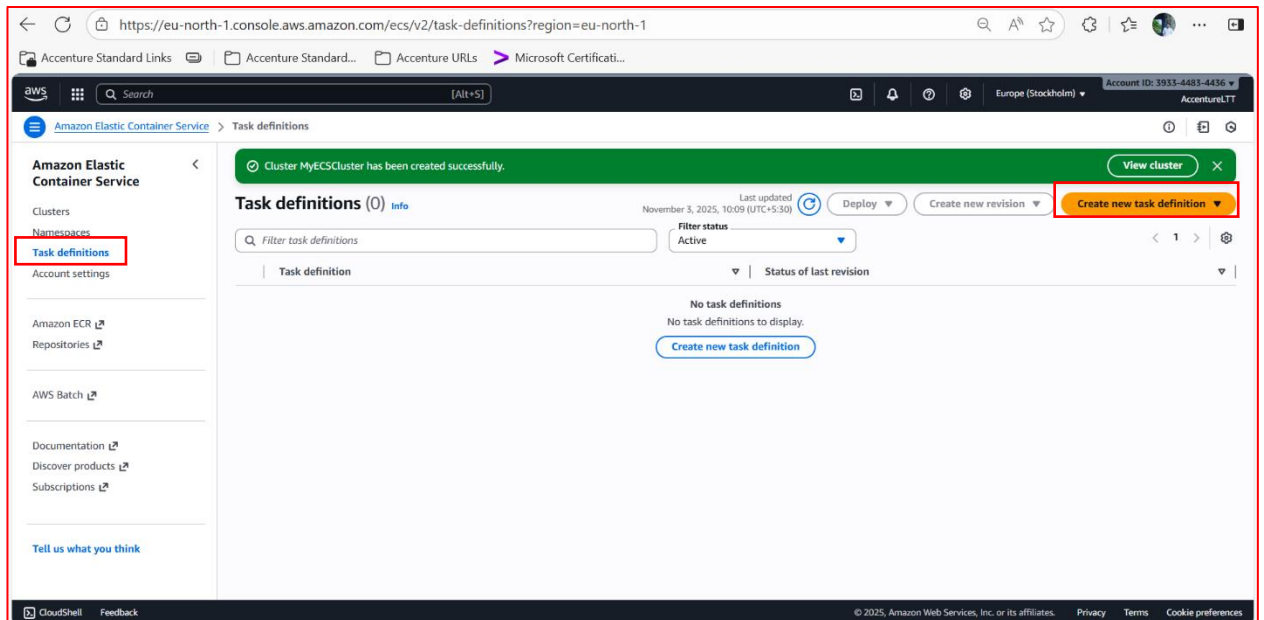
5 Amazon Elastic Container Service → Clusters → Create Cluster



- 6 Wait for some times don't refresh it may fail, wait until it gets created successfully.
You now have an empty ECS cluster.



- 7 Next Step: Create a Task Definition
A *task definition* describes what container(s) to run.



- 8 In the ECS sidebar, choose **Task definitions** → **Create new task definition**.
Select **Fargate**.
Configure:

Task name: MySampleTask

Task role: Leave default

Operating system: Linux.

Task size: 1 vCPU, 3 GB

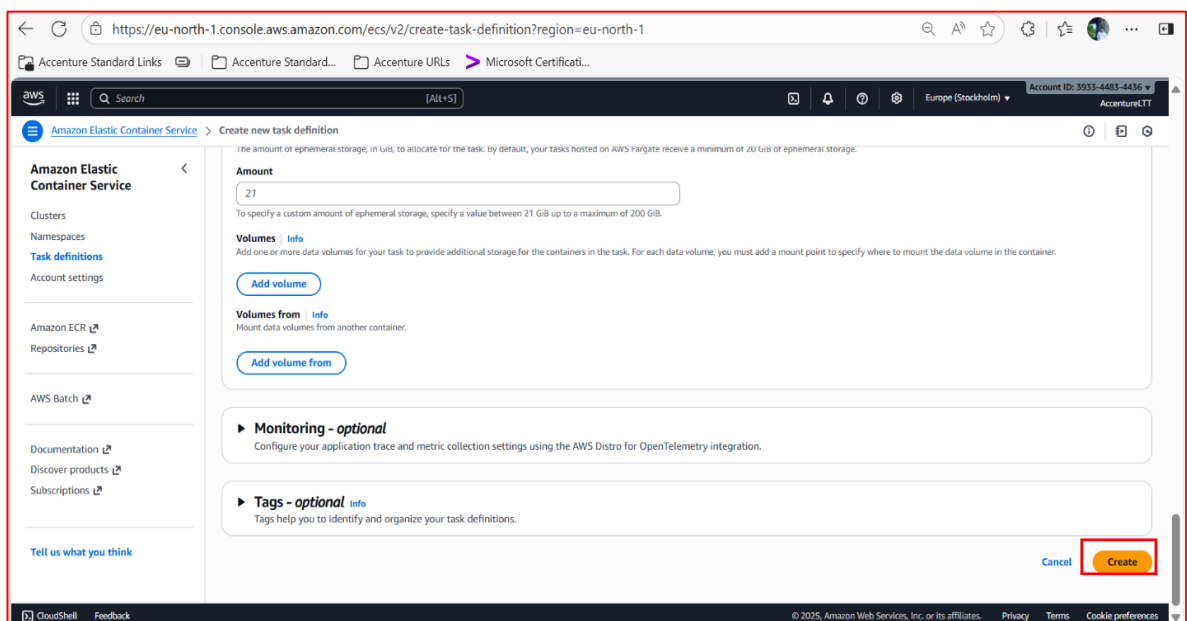
Under **Container definitions**, click **Add container**:

Container name: nginx

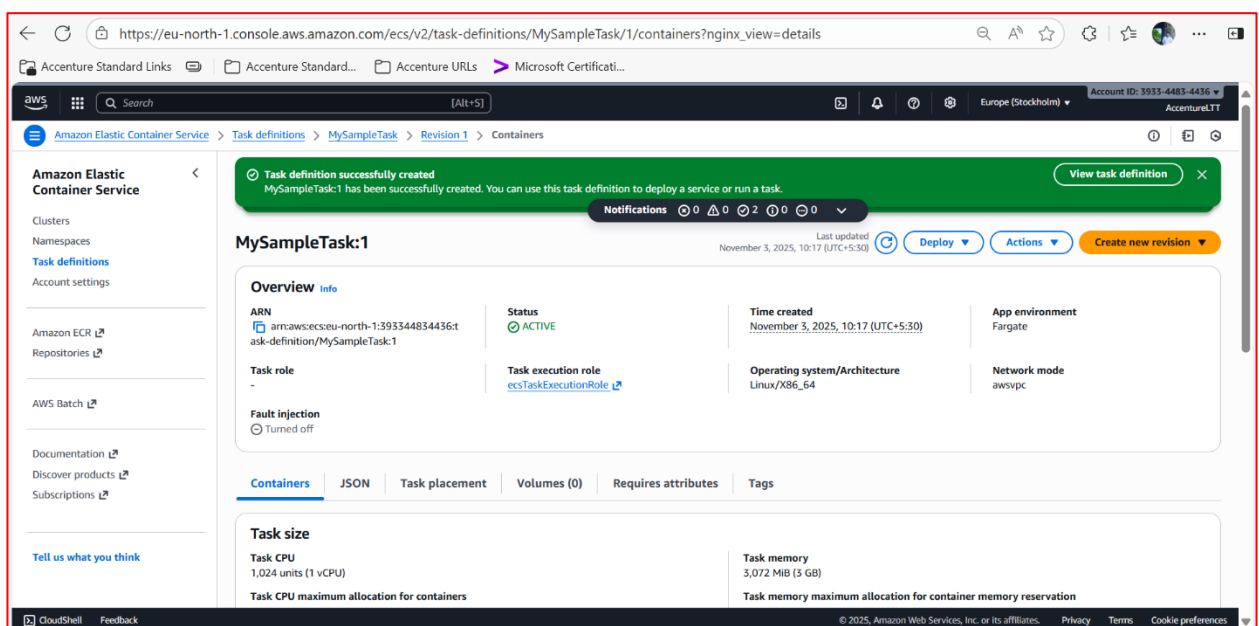
Image URI: nginx:latest (I am using available container)

Port mappings: Container port 80.

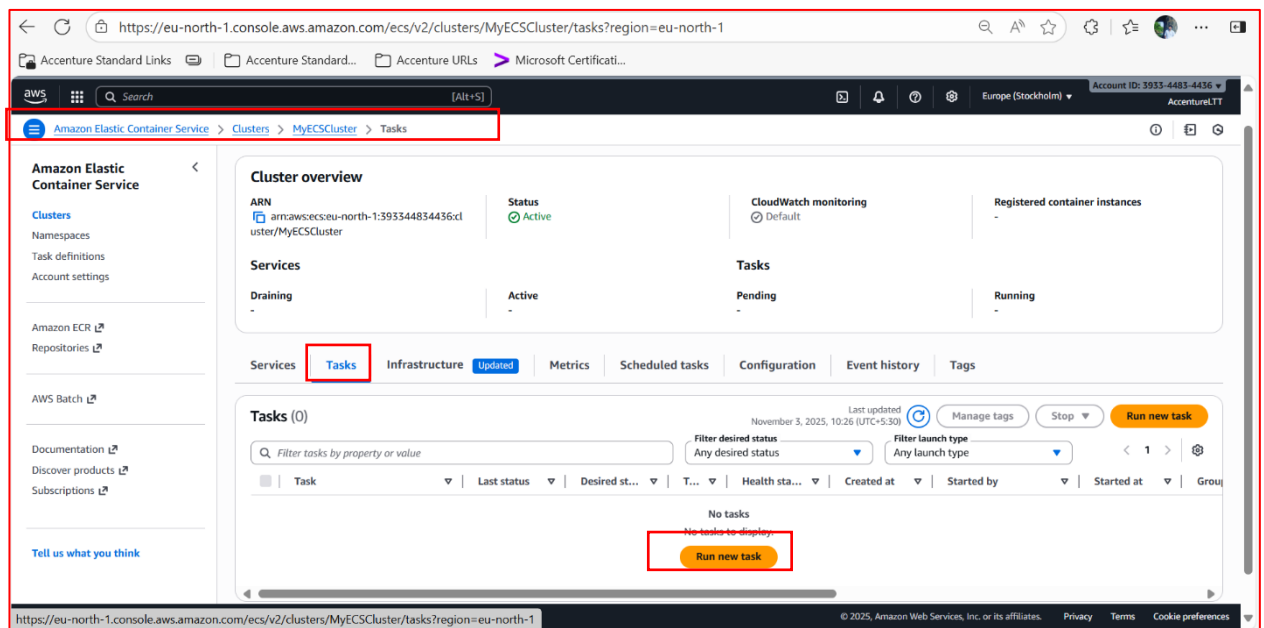
Click **Add**, then **Create**.



After Successfully Creation you can observe the status of the task created



9 Now Go to your cluster → **Tasks** → **Run new task**.

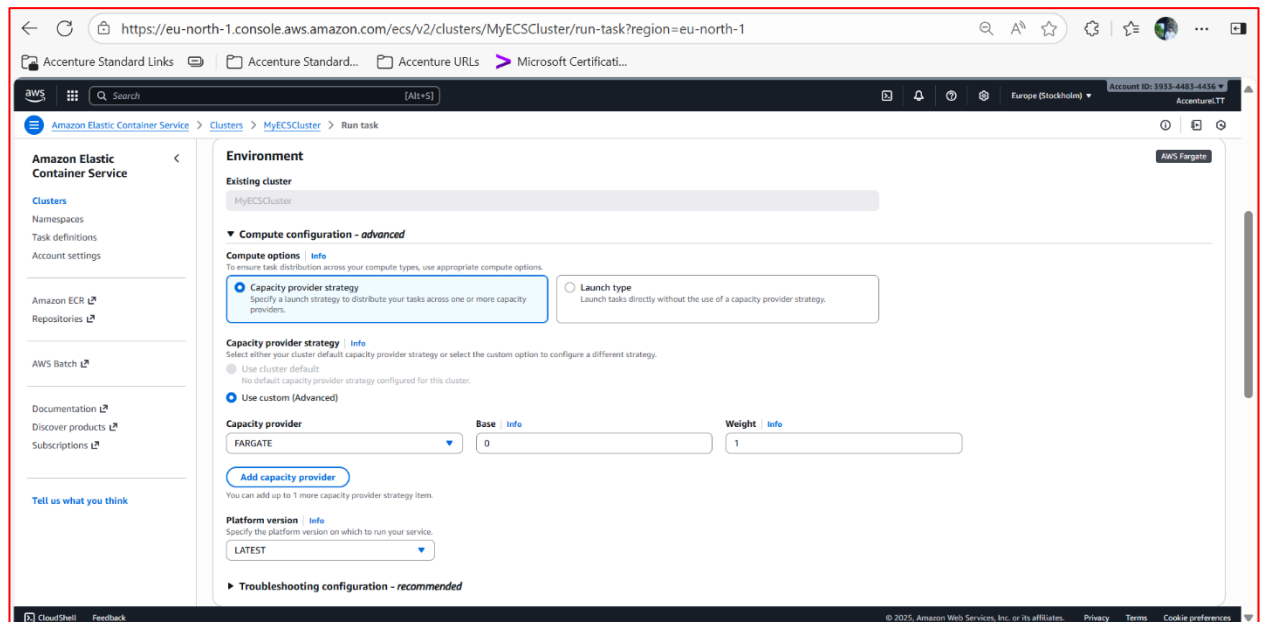


10 Choose:

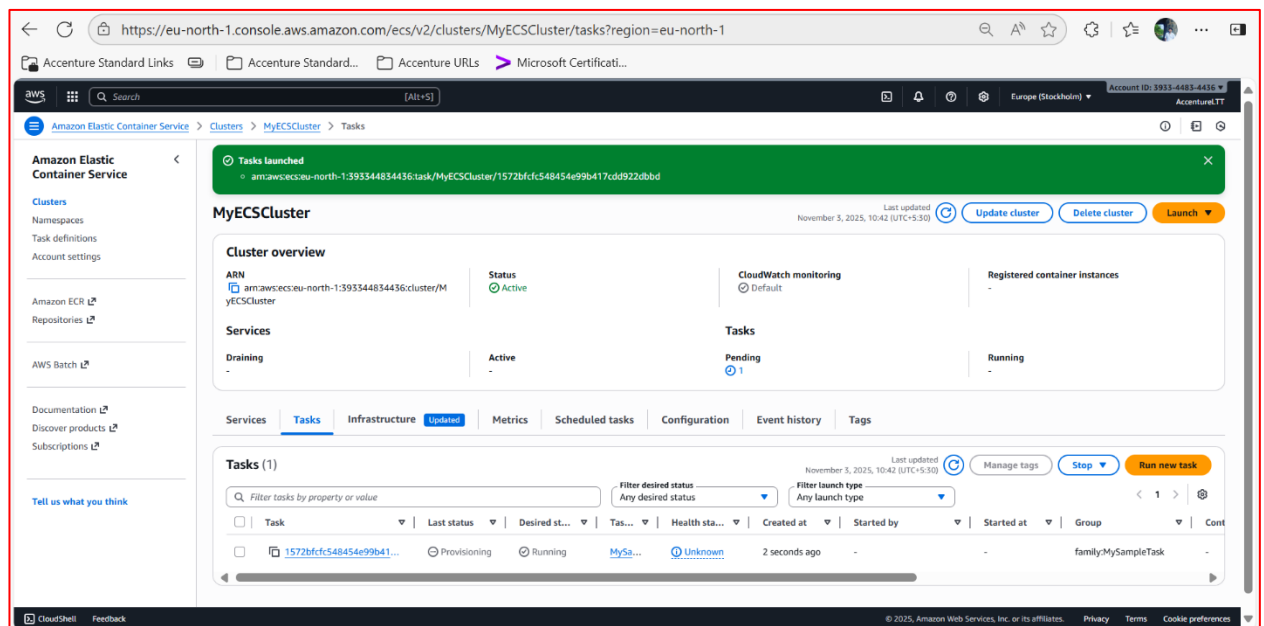
- **Launch type:** Fargate
- **Task definition:** MySampleTask
- **Cluster:** MyECSCluster
- **Number of tasks:** 1

Under **VPC and Subnets**, choose your default VPC and at least one public subnet.

Under **Security group**, use default which allows inbound traffic on port 80 (HTTP).



11 Click Create, AWS will now launch a container in Fargate

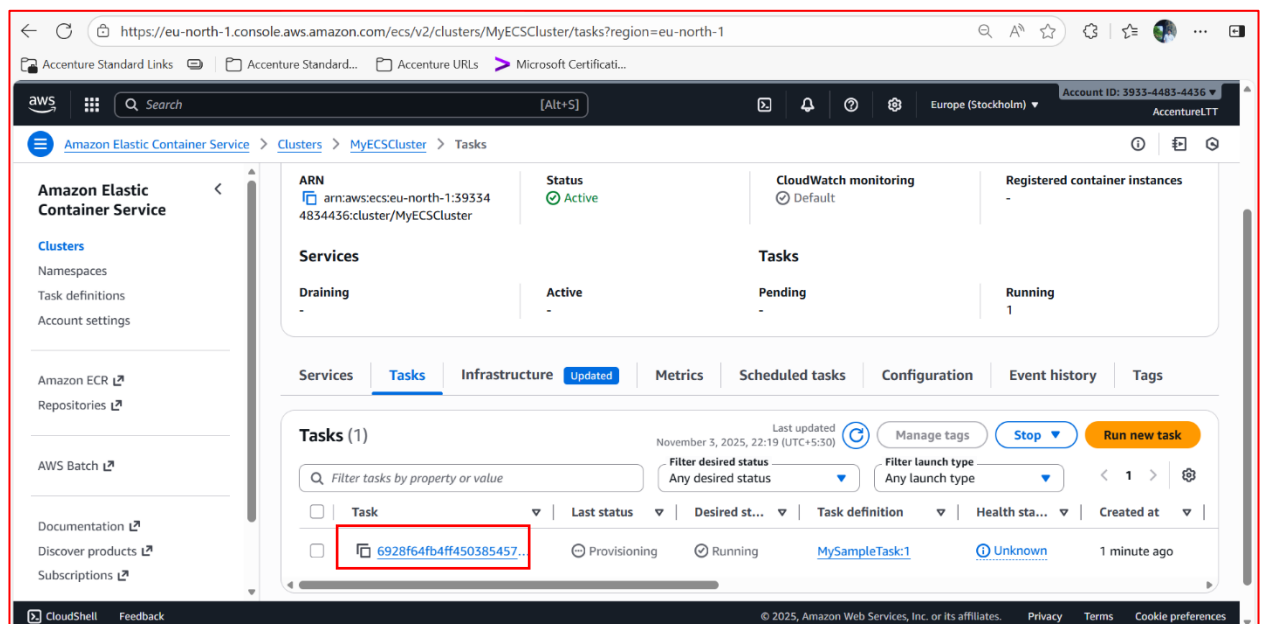


12 Check It's Running

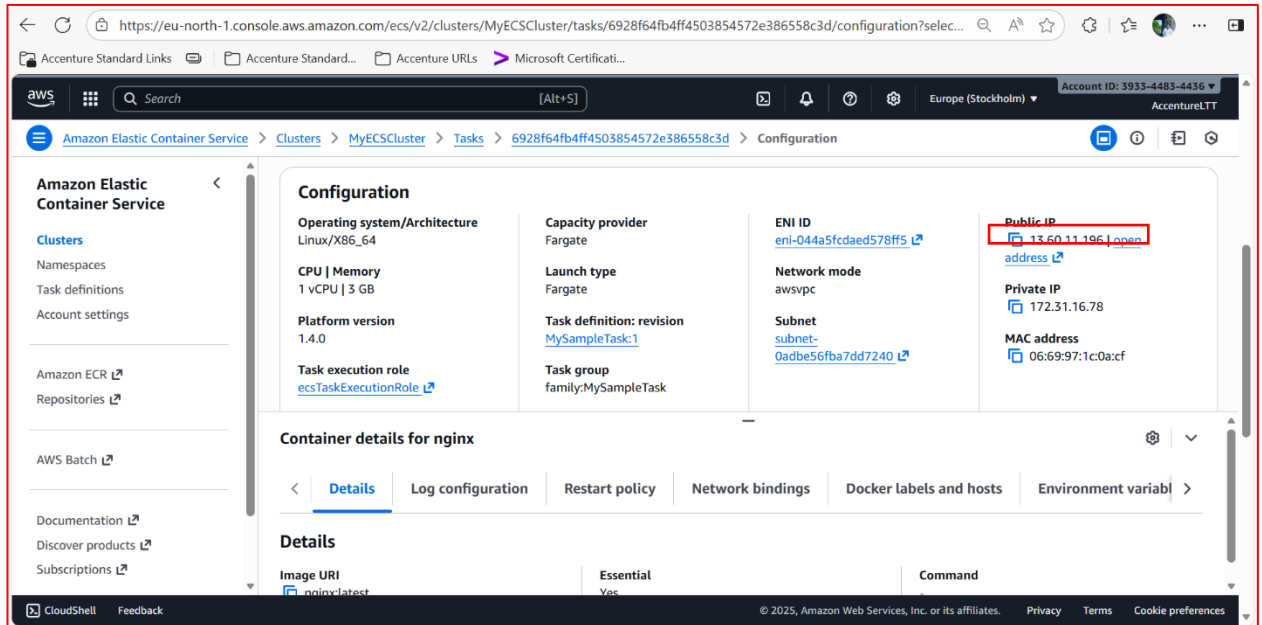
Go to your cluster → Tasks → [task ID].

Find the ENI (Elastic Network Interface) or Public IP address.

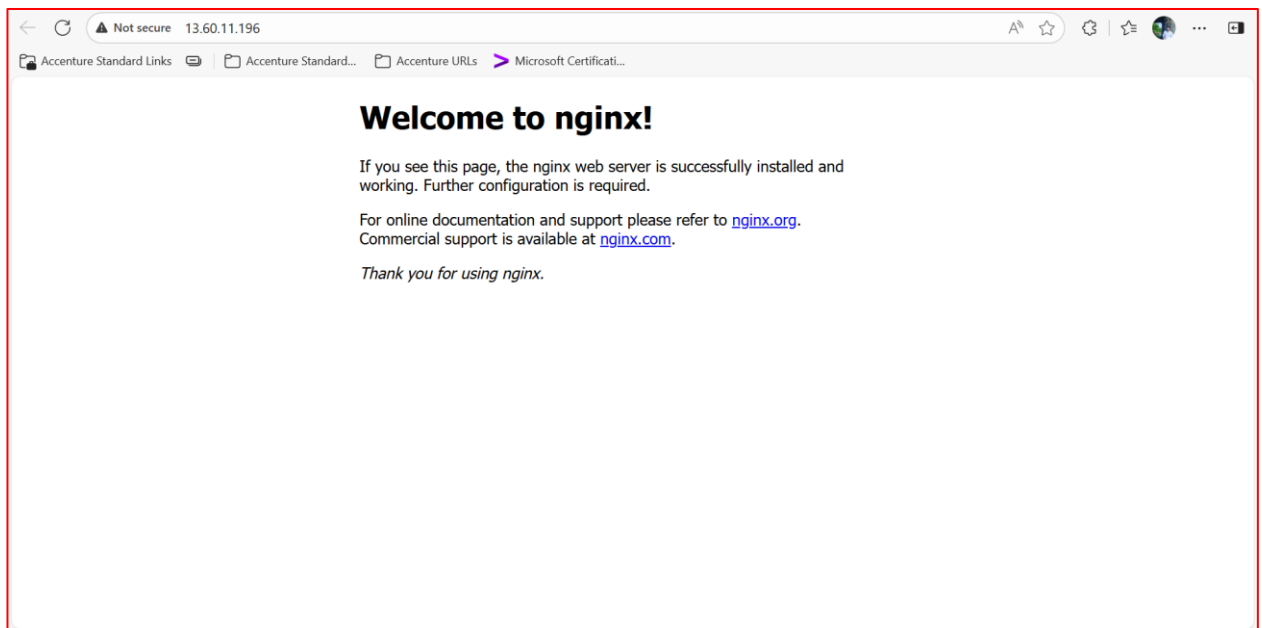
Open that IP in your browser → you should see the default Nginx welcome page.



- 13 After clicking on Created Task
You can observe the public IP address → Click on IP address



- 14 Default Nginx welcome page will be launched.



15 Once the tasks are completed, please stop, so you can observe the task status as Stopped

The screenshot shows the AWS Management Console interface for the Amazon Elastic Container Service (ECS). The breadcrumb navigation at the top indicates the path: **Amazon Elastic Container Service** > **Clusters** > **MyECSCluster** > **Tasks**. The left-hand navigation pane lists various ECS-related options: **Clusters**, **Namespaces**, **Task definitions**, **Account settings**, **Amazon ECR**, **Repositories**, **AWS Batch**, **Documentation**, **Discover products**, and **Subscriptions**. The main content area displays the 'Tasks' page for the cluster '4834436:cluster/MyECSCluster'. At the top, there are tabs for **Services**, **Tasks** (which is selected), **Infrastructure**, **Updated**, **Metrics**, **Scheduled tasks**, **Configuration**, **Event history**, and **Tags**. Below these tabs, the 'Tasks (1)' section shows a table of tasks. The table has columns for **Task**, **Last status**, **Desired st...**, **Task definition**, **Health sta...**, and **Created at**. A single task is listed with ID '6928f64fb4ff450385457...', status 'Stopped', and task definition 'MySampleTask:1'. The 'Stop' button in the top right corner of the task list is highlighted with a red box. The 'Stopped' status in the 'Last status' column is also highlighted with a red box.

4. Activity to create EKS

1

Prerequisites (IAM Roles and Networking)

Before creating the cluster, you need foundational resources.

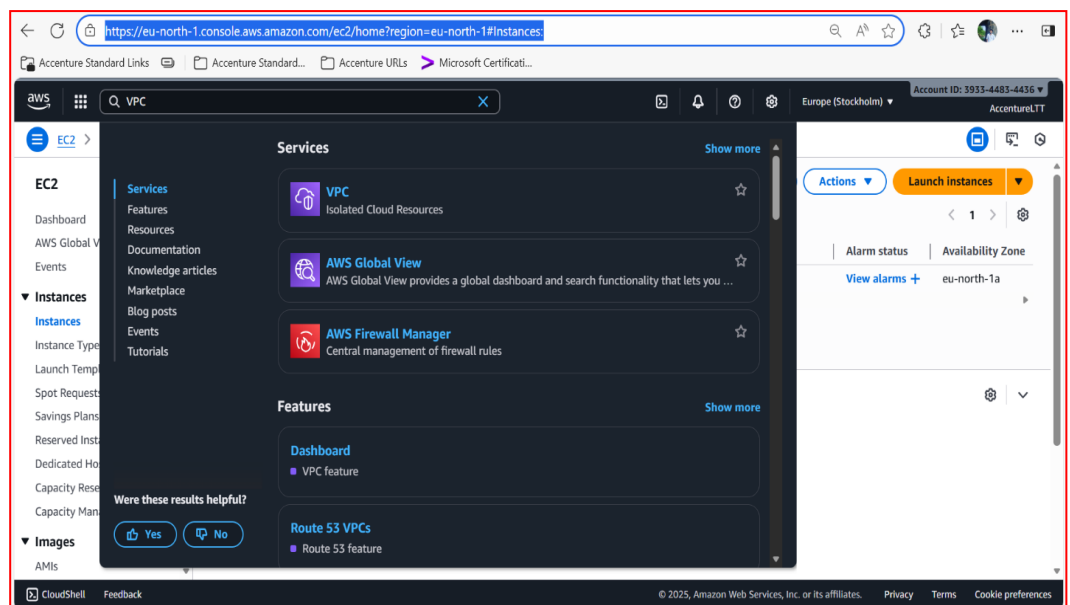
Create a VPC and Subnets

EKS requires a Virtual Private Cloud (VPC) with subnets across at least two Availability Zones (AZs).

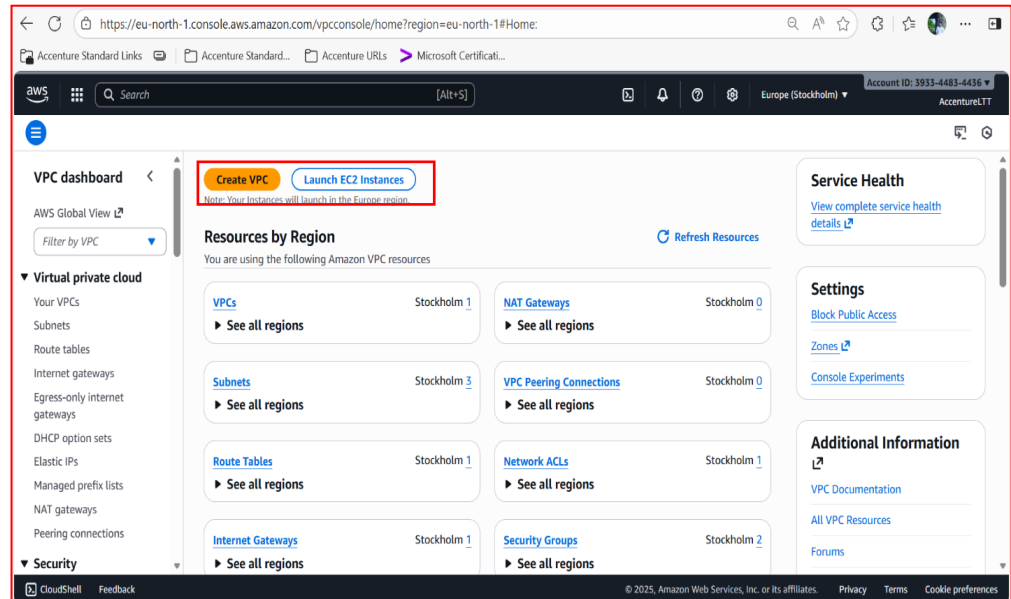
The easiest way to do this is using the VPC Wizard:

Sign In to AWS

Search for VPC in search bar



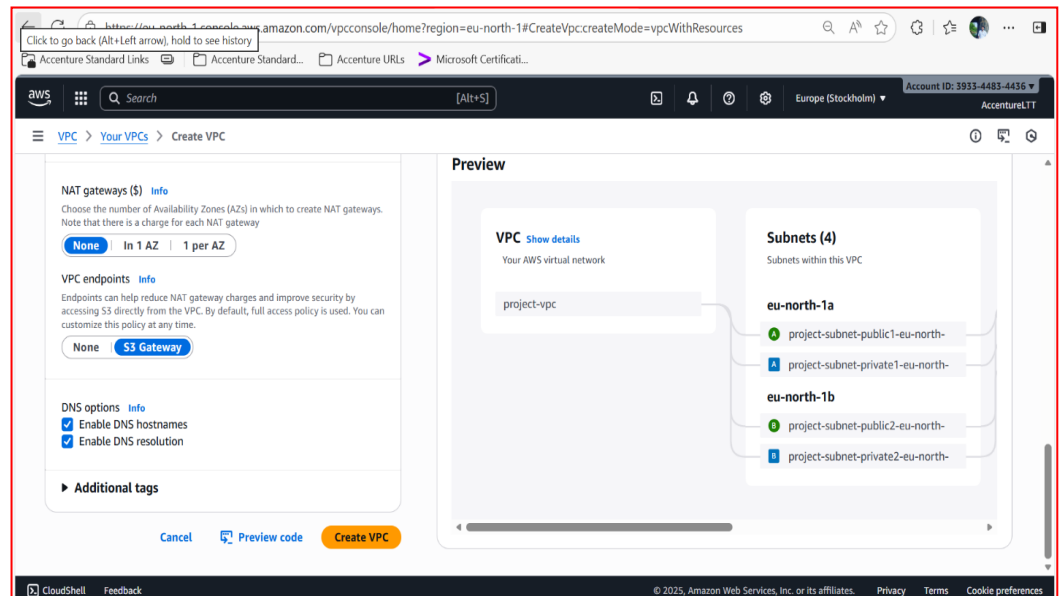
2

Go to the **VPC** console.

3

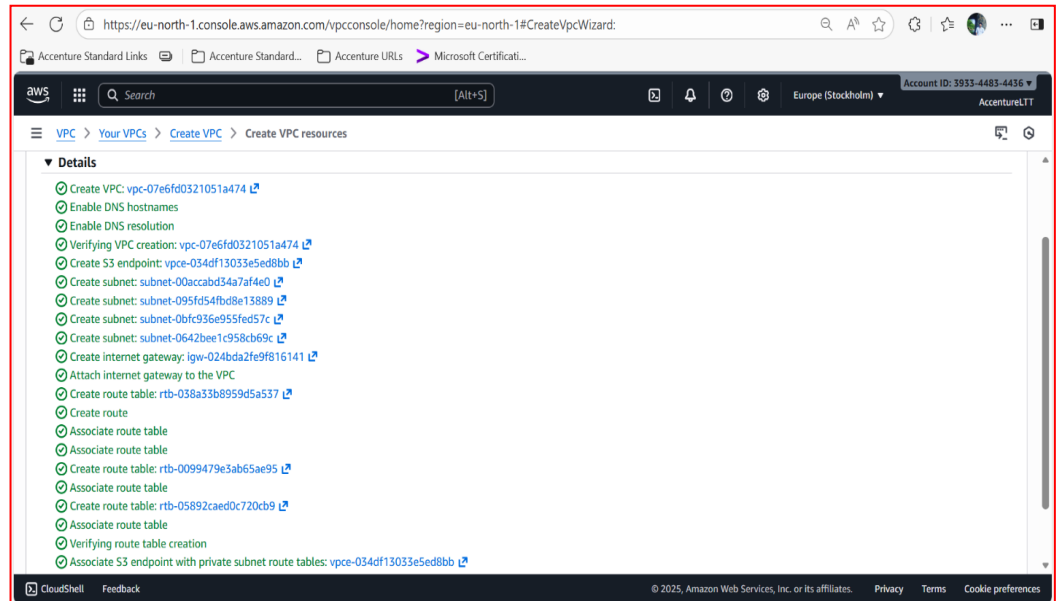
In the left navigation pane, choose **Your VPCs**, then click **Create VPC**.

1. Select **VPC and more** (This is the fastest method).
2. For **Number of Availability Zones (AZs)**, choose **2**.
3. Keep the defaults for subnets (2 public, 2 private).
4. Click **Create VPC**. This will create a VPC, subnets, Internet Gateway, and NAT Gateway (which costs money but is needed for worker nodes in private subnets to pull images). **Note down the VPC ID.**



4

View VPC once created successfully.

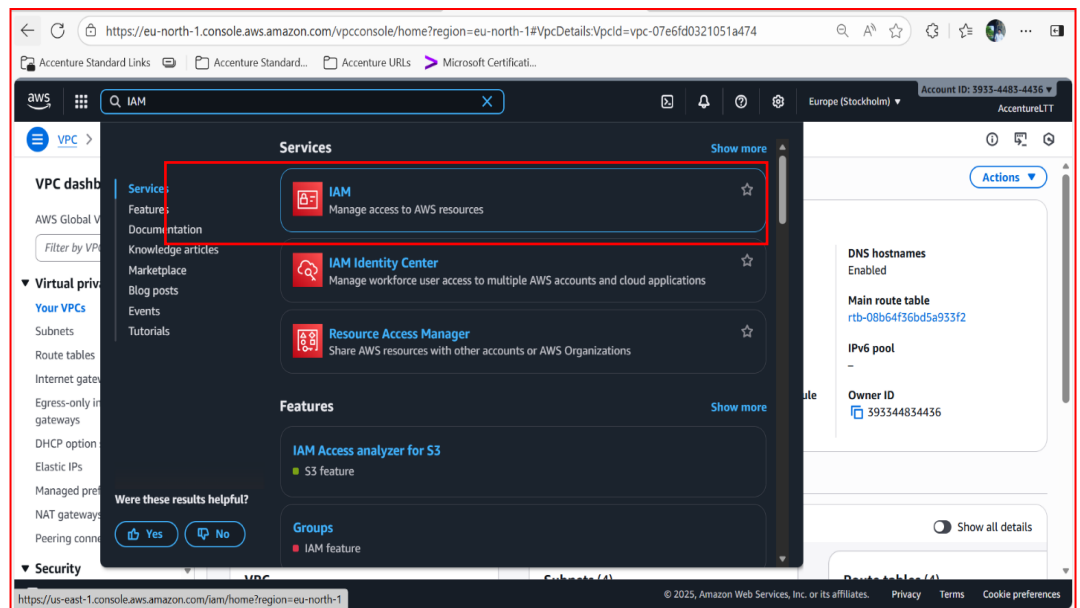


5

Create an EKS Cluster IAM Role

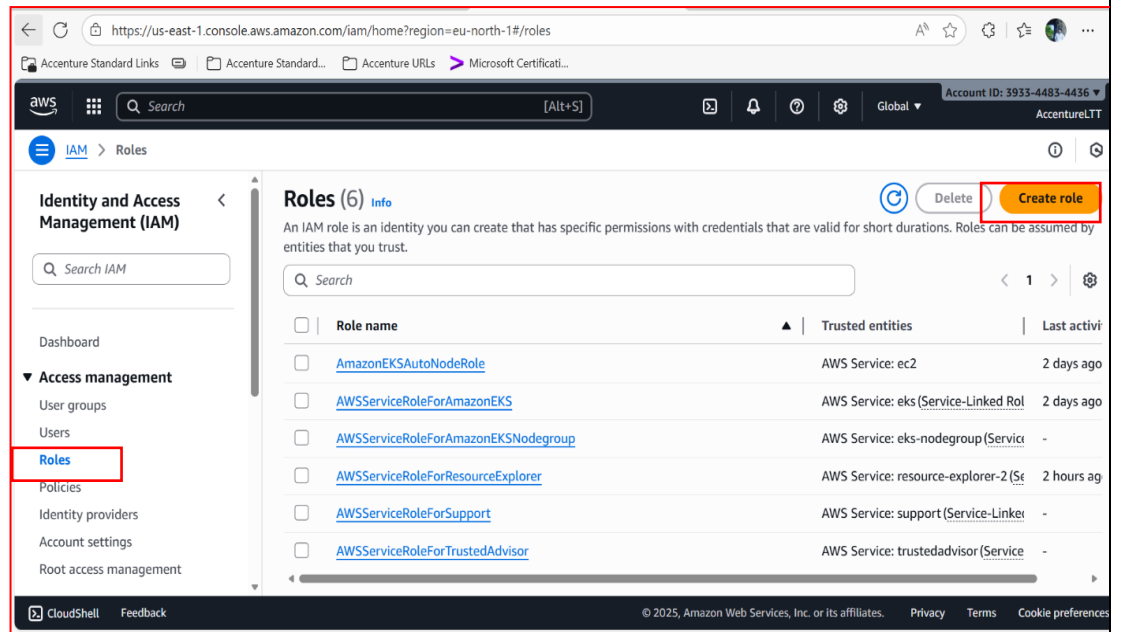
The EKS control plane needs permission to manage AWS resources (like EC2 instances, VPC, and Load Balancers) on your behalf.

Go to the **IAM** console.



6

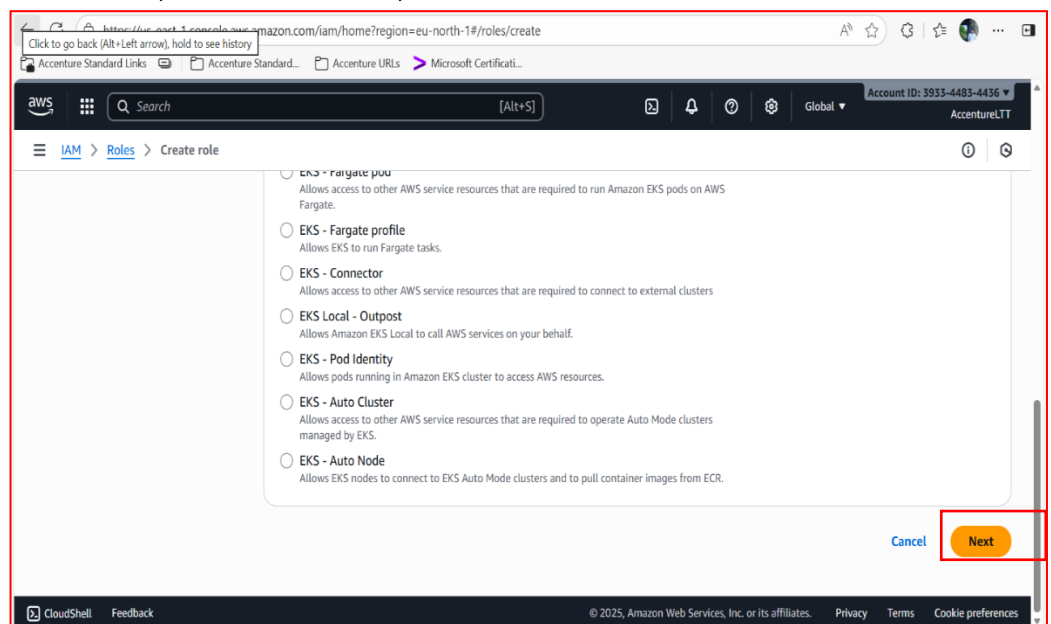
Choose **Roles**, then click **Create role**.



7

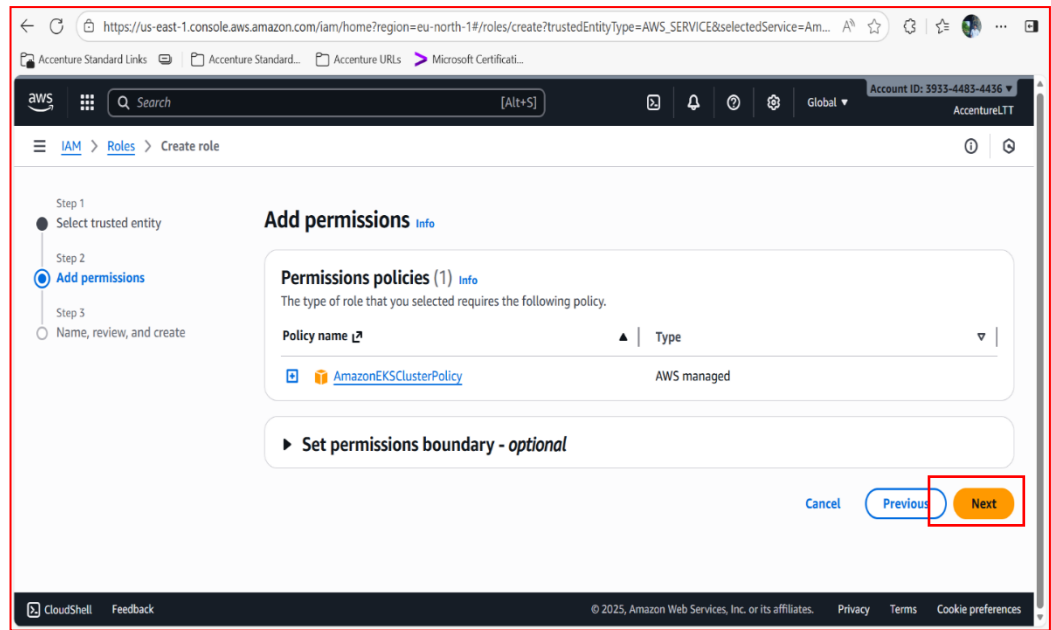
For **Trusted entity type**, select **AWS service**.

For **Use case**, find and select **EKS**, and then choose the **EKS - Cluster** sub-use case.



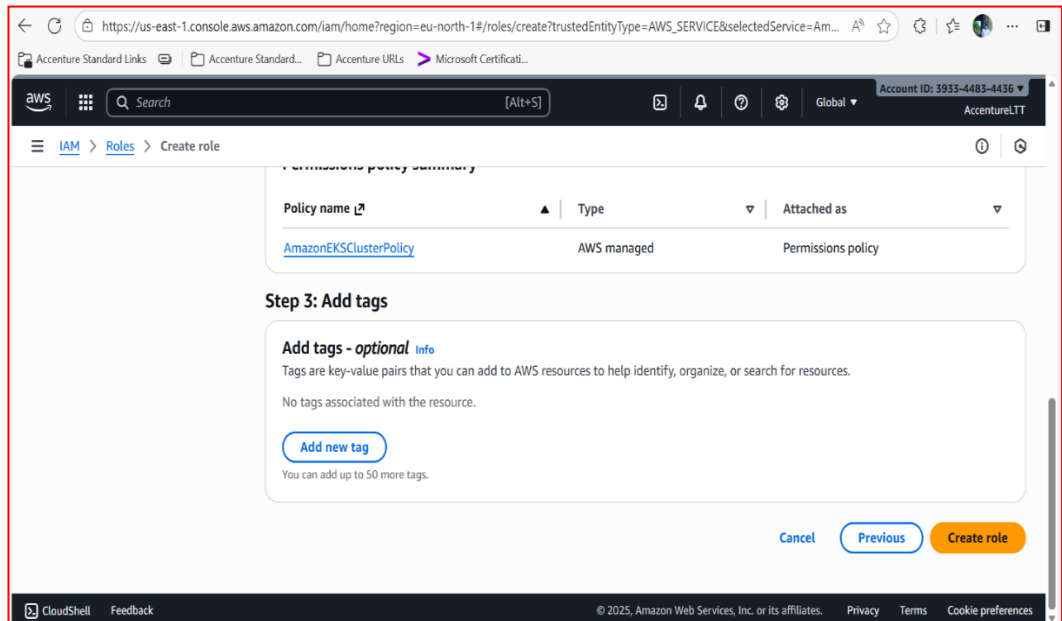
8

Click **Next**. The AmazonEKSClusterPolicy will be pre-selected.

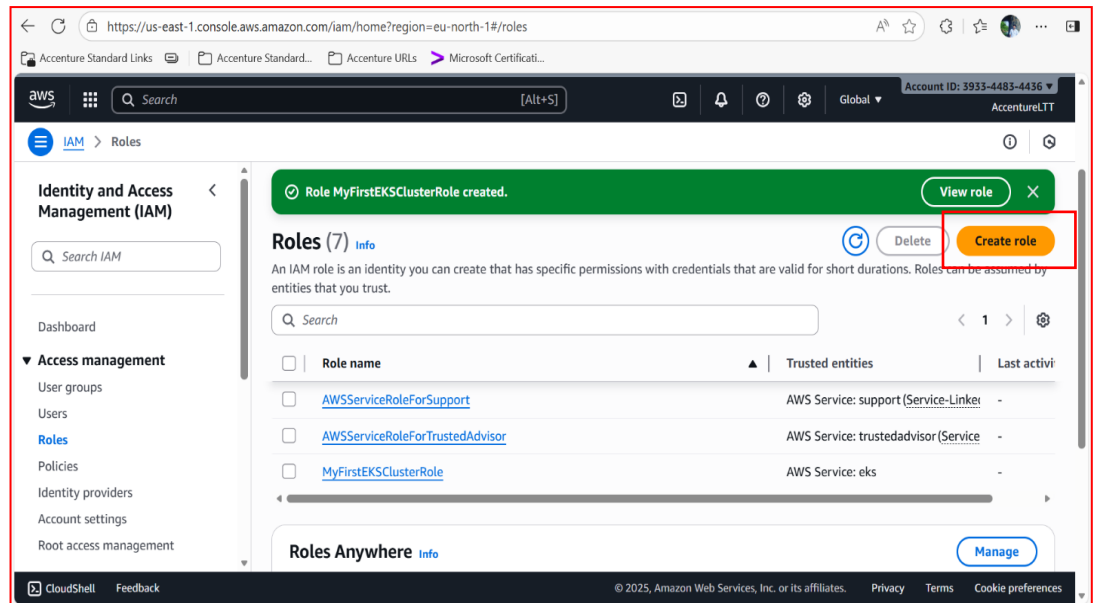


9

Click **Next**, then provide a **Role name** (e.g., MyFirstEKSClusterRole).

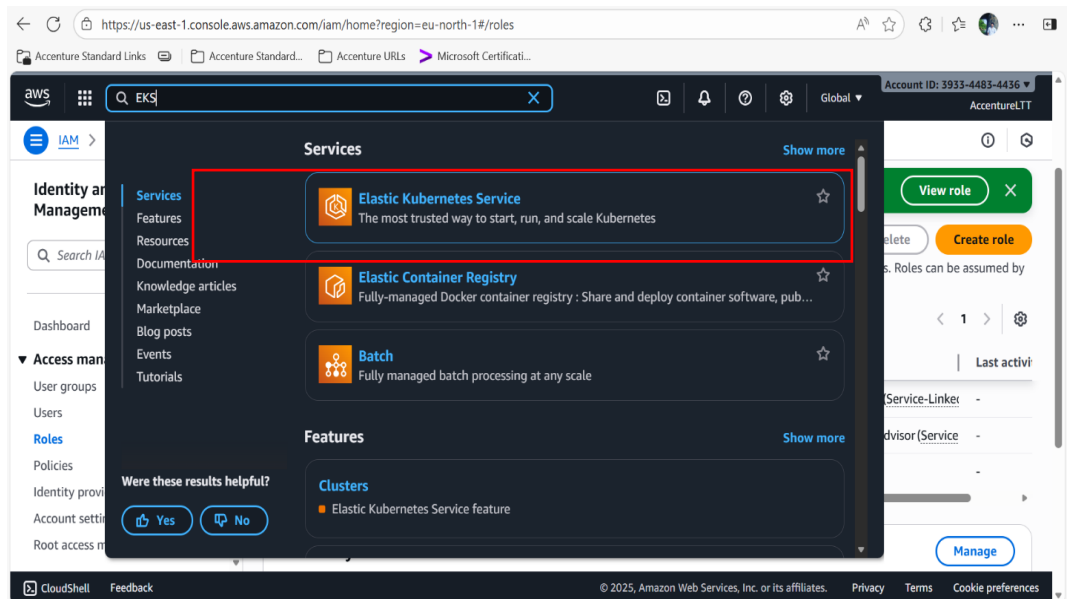


10

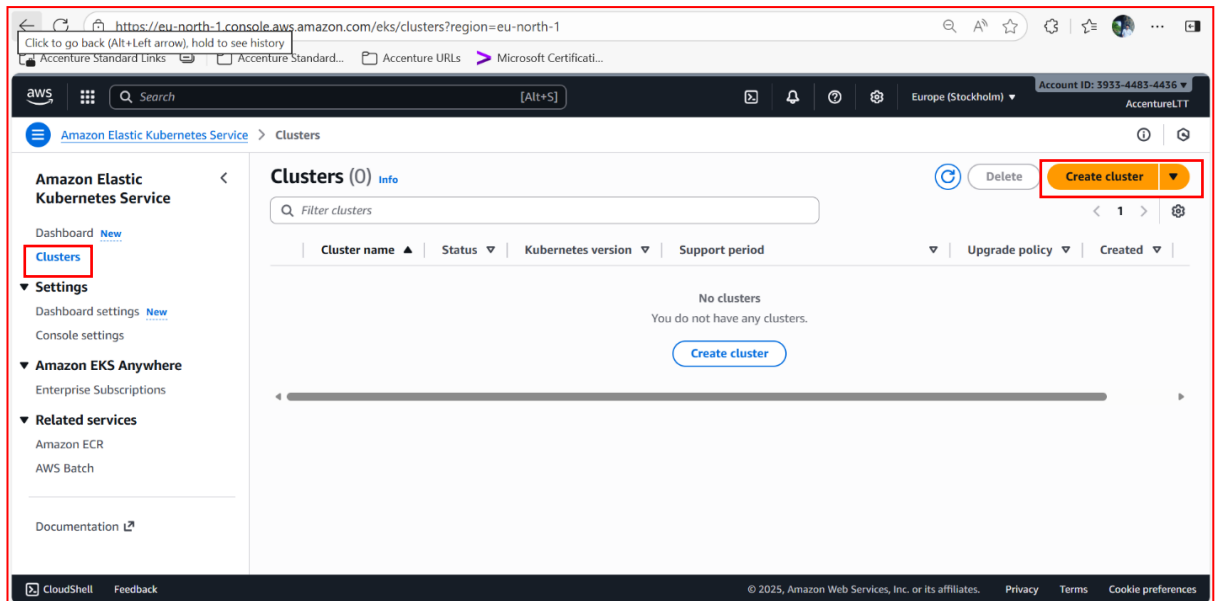
Click **Create role**.

11

Create the EKS Cluster Control Plane
Go to the **Amazon EKS** console.



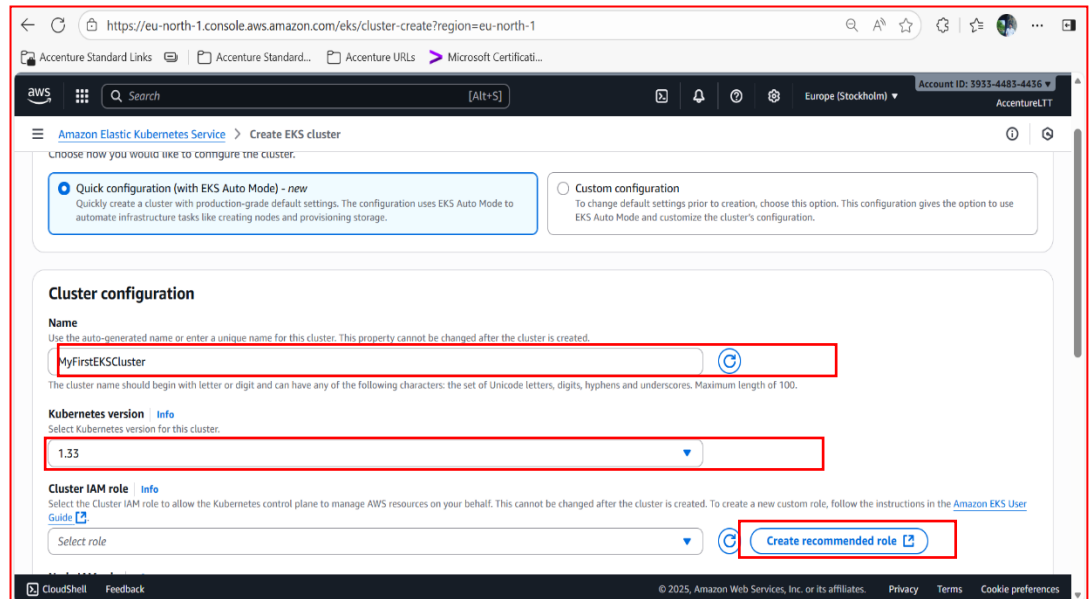
12

Click the **Clusters** link in the left pane, then click **Create cluster**.

13

Configure cluster:

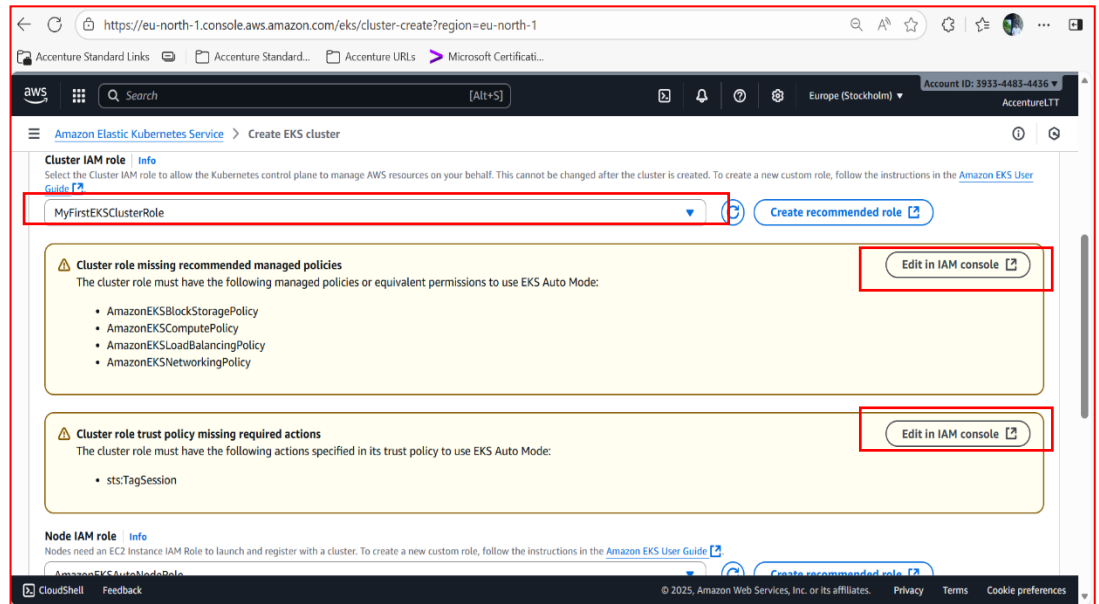
Name: Enter a unique name (e.g., MyFirstEKSCluster).



14

Kubernetes version: Choose the latest supported version.

Cluster service role: Select the **Cluster IAM Role** you created in previous step (e.g., MyFirstEKSClusterRole).



15

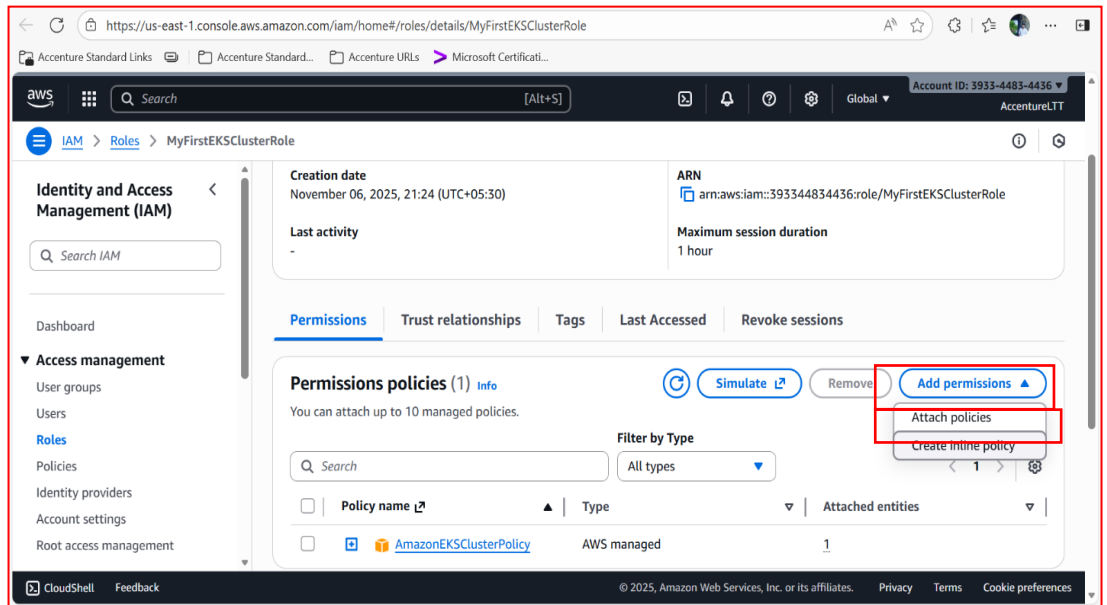
Update the Cluster Role for EKS Auto Mode

The simplest way to fix this is to use the AWS Console's recommendation feature.

1. **Click the "Create recommended role" button:** In the top right corner of the image you provided, click the **Create recommended role** button.
 - If you're still on the EKS cluster creation page, this button should automatically create a new, compliant IAM role and select it for your cluster.
2. **Manually Edit the IAM Role (If you click "Edit in IAM console"):**
 - If you choose to use your existing role (MyFirstEKSClusterRole), you need to navigate to the **IAM Console** and attach the missing policies and update the trust policy.
3. **Attach Missing Managed Policies**

You must attach the four policies listed in the first warning:

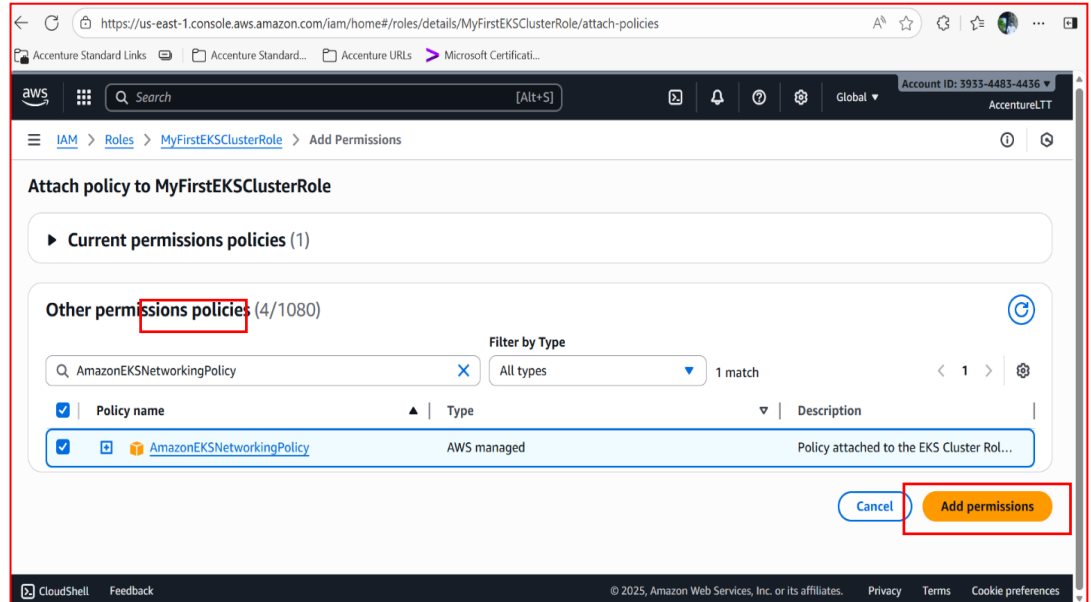
 1. Go to the **IAM Console** -> **Roles**.
 2. Search for and select your cluster role (MyFirstEKSClusterRole).
 3. Click **Add permissions** -> **Attach policies**.



16

4. Search for and attach the following four AWS Managed Policies:

- AmazonEKSBLOCKStoragePolicy
- AmazonEKSCOMPUTEPolicy
- AmazonEKSLoadBalancingPolicy
- AmazonEKSNetworkingPolicy

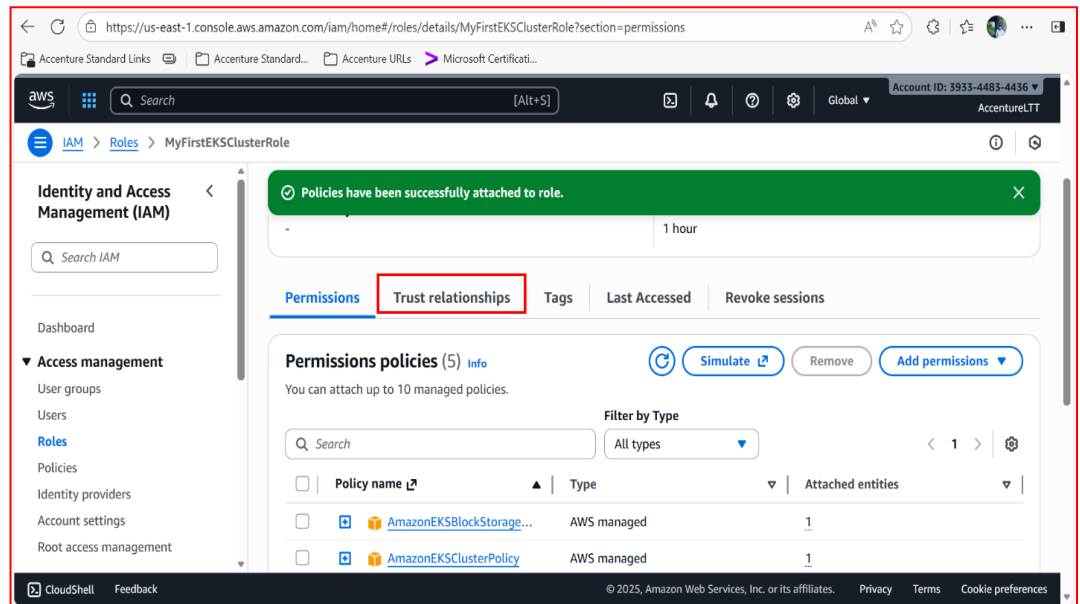


17

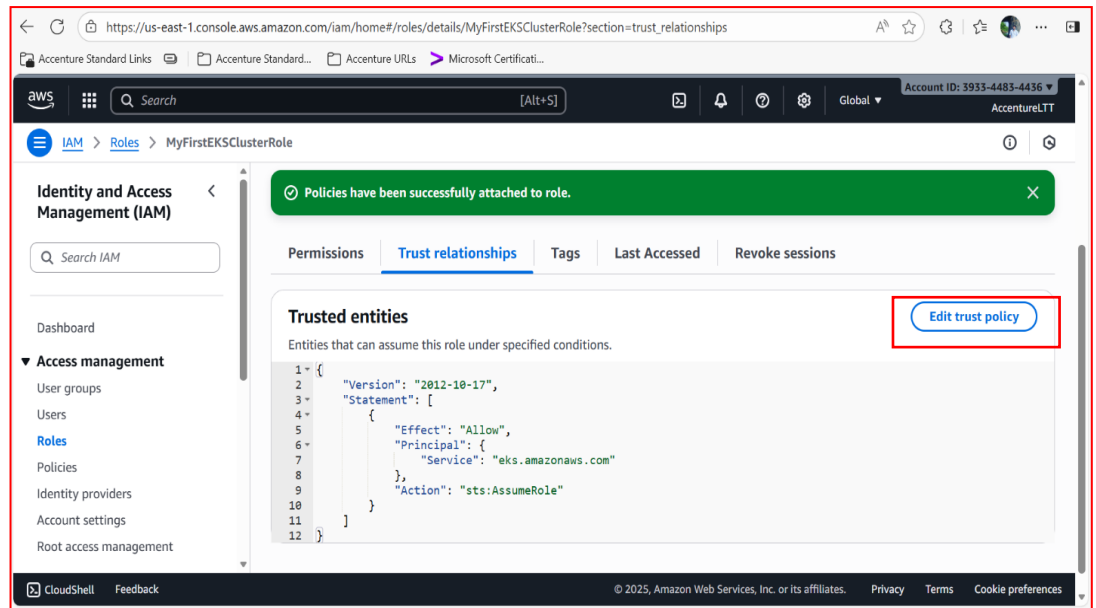
Update the Trust Relationship

You must add the sts:TagSession action to the trust policy as indicated in the second warning.

1. While still viewing your cluster role (MyFirstEKSClusterRole), click the **Trust relationships** tab.

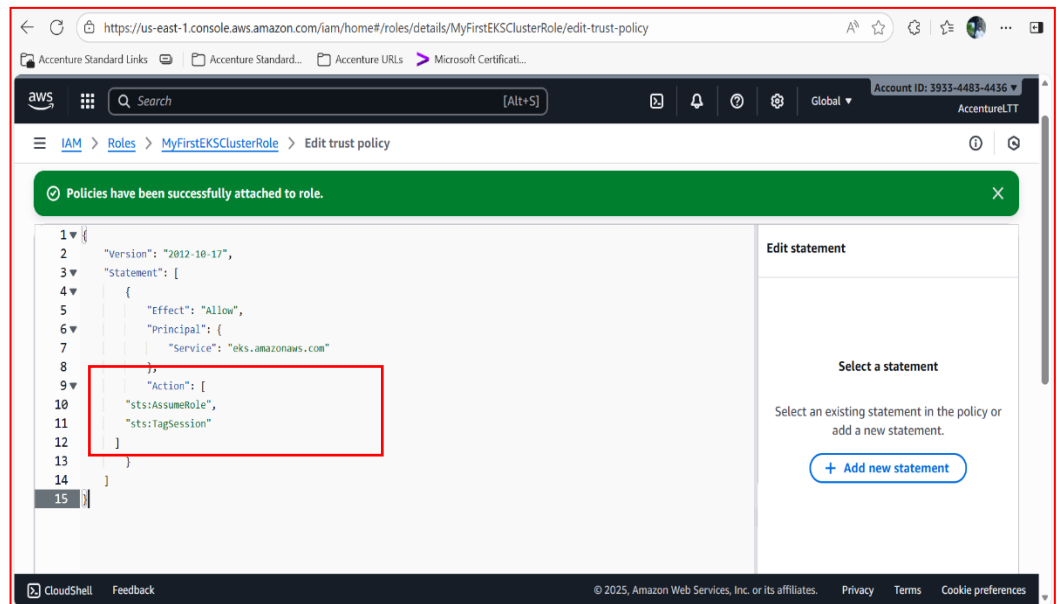


18

2. Click **Edit trust policy**.

19

3. Locate the existing statement that allows the EKS service to assume the role (where "Service": "eks.amazonaws.com").
4. Add "sts:TagSession" to the Action array.



20

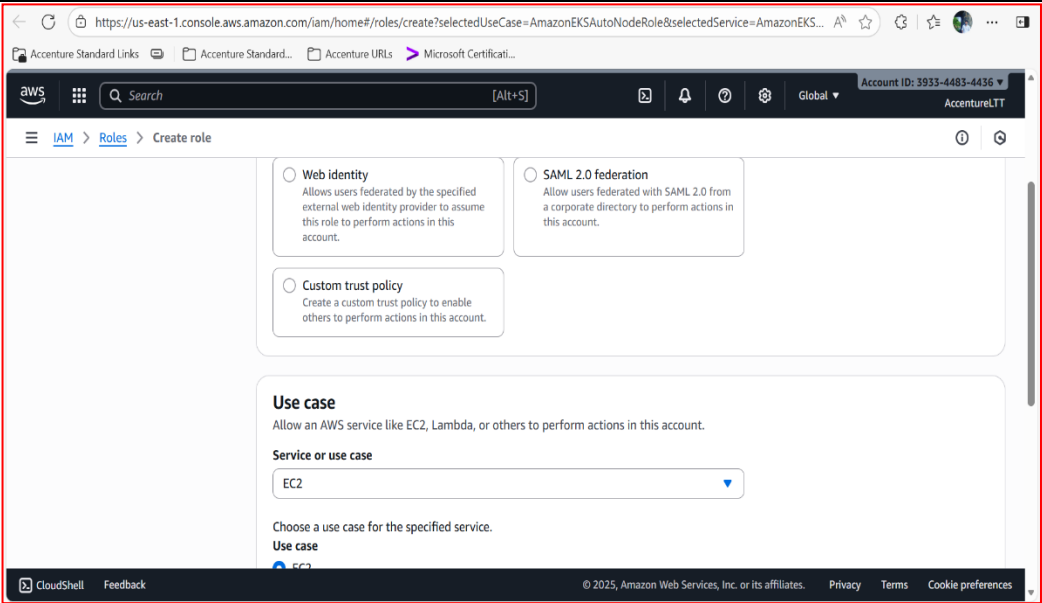
Click **Update Policy**.

Once you have either created the recommended role or manually updated your existing role, click the **refresh icon** next to the role dropdown in the EKS console.

15

Create the Node IAM Role

1. Navigate to the **IAM console** in a new browser tab.
2. In the left navigation pane, choose **Roles**, and then click **Create role**.
3. For **Trusted entity type**, select **AWS service**.
4. For the **Use case**, select **EC2** (because your nodes are essentially EC2 instances).
5. Click **Next**.



16

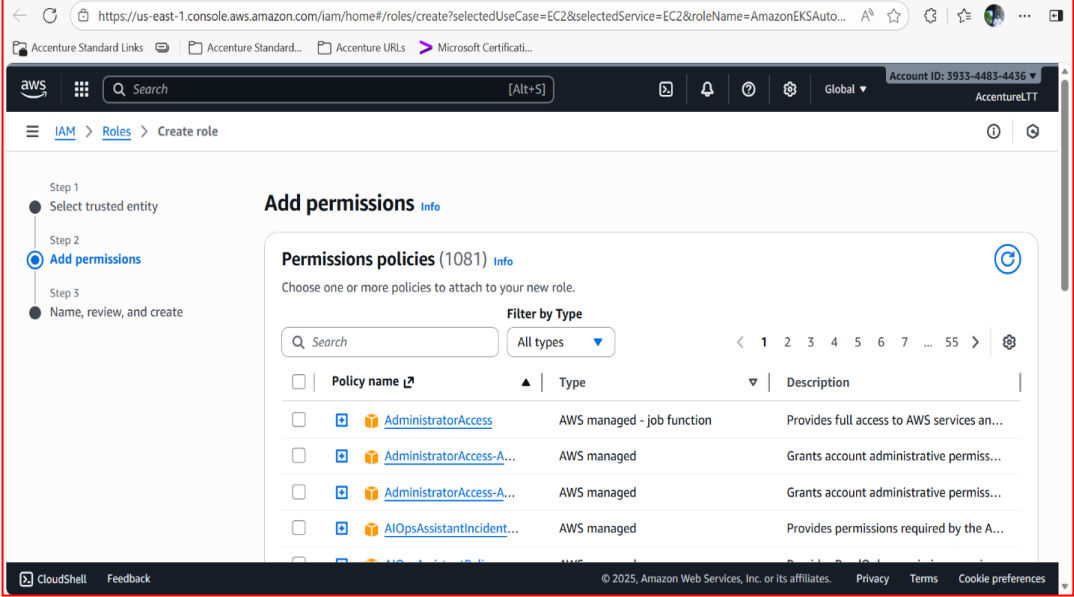
Attach Required Policies

On the "Add permissions" page, you must attach the following three AWS Managed Policies:

AmazonEKSWorkerNodePolicy

AmazonEKS_CNI_Policy

AmazonEC2ContainerRegistryReadOnly



Add Required Policies

https://us-east-1.console.aws.amazon.com/iam/home#/roles/create?selectedUseCase=EC2&selectedService=EC2&roleName=AmazonEKSAuto...

Account ID: 3933-4483-4436

IAM > Roles > Create role

permissions policy summary

Policy name	Type	Attached as
-------------	------	-------------

Step 3: Add tags

Add tags - optional [info](#)

Tags are key-value pairs that you can add to AWS resources to help identify, organize, or search for resources.

No tags associated with the resource.

[Add new tag](#)

You can add up to 50 more tags.

[Cancel](#) [Previous](#) [Create role](#)

Create role

Once you have either created the recommended role or manually updated your existing role, click the **refresh icon** next to the role dropdown in the EKS console.

https://eu-north-1.console.aws.amazon.com/eks/cluster-create?region=eu-north-1

Account ID: 3933-4483-4436

Europe (Stockholm)

Amazon Elastic Kubernetes Service > Create EKS cluster

Cluster IAM role

Select the Cluster IAM role to allow the Kubernetes control plane to manage AWS resources on your behalf. This cannot be changed after the cluster is created. To create a new custom role, follow the instructions in the [Amazon EKS User Guide](#).

[Guide](#)

MyFirstEKSClusterRole [Create recommended role](#)

Node IAM role

Nodes need an EC2 Instance IAM Role to launch and register with a cluster. To create a new custom role, follow the instructions in the [Amazon EKS User Guide](#).

[Guide](#)

AmazonEKSAutoNodeRole [Create recommended role](#)

VPC

Select a VPC to use for your EKS cluster resources.

vpc-07e6fd0321051a474 | project-vpc [Create VPC](#)

Subnets

Choose the subnets in your VPC where the control plane may place elastic network interfaces (ENIs) to facilitate communication with your cluster. To create a new subnet, go to the corresponding page in the VPC console.

[Select subnets](#) [Clear selected subnets](#)

subnet-0bfc936e955fed57c | project-subnet-private1-eu-north-1a [X](#)

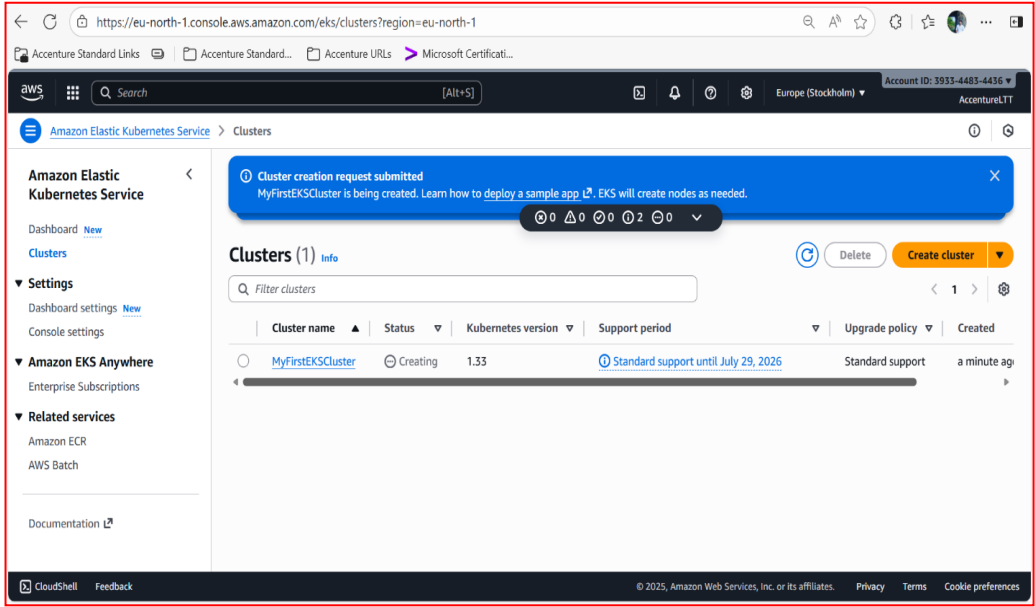
eu-north-1a 10.0.128.0/20 Type: Private

subnet-0642bee1c958cb69c | project-subnet-private2-eu-north-1b [X](#)

eu-north-1b 10.0.144.0/20 Type: Private

18

Select VPC Created in previous step
Click on Create



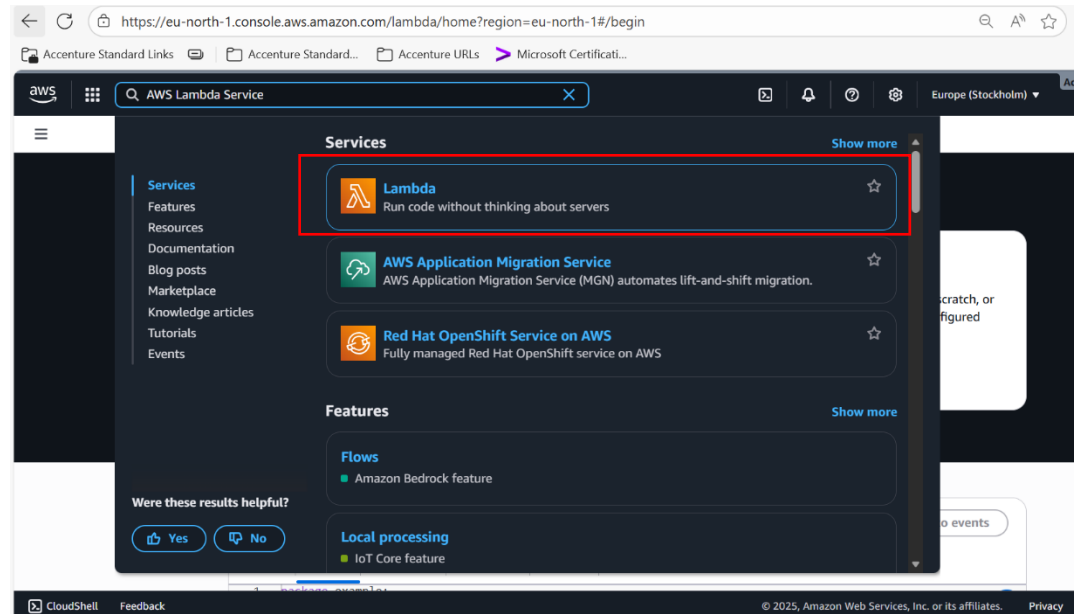
Your Cluster will be created
Thank You!

5. Create the Lambda Function (Console)

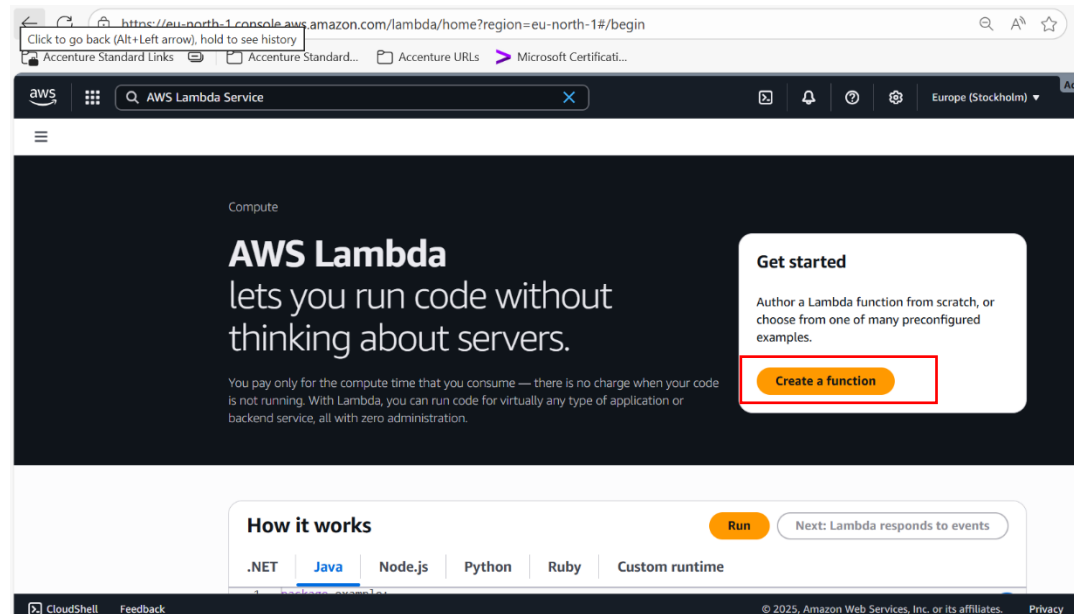
1 Sign in to the AWS Management Console.

[Console Home](#) | [Console Home](#) | [eu-north-1](#)

Navigate to the **AWS Lambda** service.



2 Click **Create function**.



3

1. Choose **Author from scratch**.
2. **Basic Information:**
 1. **Function name:** eg., MyFirstLambdaFunction.
 2. **Runtime:** Select your desired programming language (e.g., **Python 3.12, Node.js 20**). This defines the execution environment.
 3. **Architecture:** Choose **x86_64** or **arm64** (arm64, powered by AWS Graviton2, often provides better performance and lower cost).
 4. **Permissions:** For the **Execution role**, choose **Create a new role with basic Lambda permissions**. This creates a role allowing the function to write logs to **Amazon CloudWatch**.

https://eu-north-1.console.aws.amazon.com/lambda/home?region=eu-north-1#/create/function?firstrun=true&intent=authorFromScratch

Accenture Standard Links | Accenture Standard... | Accenture URLs | Microsoft Certificati...

aws | AWS Lambda Service | Europe (Stockholm)

Lambda > Functions > Create function

Basic information

Function name
Enter a name that describes the purpose of your function.

Function name must be 1 to 64 characters, must be unique to the Region, and can't include spaces. Valid characters are a-z, A-Z, 0-9, hyphens (-), and underscores (_).

Runtime [Info](#)
Choose the language to use to write your function. Note that the console code editor supports only Node.js, Python, and Ruby.

Architecture [Info](#)
Choose the instruction set architecture you want for your function code.
☐ arm64
☒ x86_64

Permissions [Info](#)
By default, Lambda will create an execution role with permissions to upload logs to Amazon CloudWatch Logs. You can customize this default role later when adding triggers.

[Change default execution role](#)

CloudShell | Feedback | © 2025, Amazon Web Services, Inc. or its affiliates. | Privacy

Info
Learn how to use cases in

Create a new app
In this tutorial, how to:

- Build a container function
- Upload code to the function
- Invoke the function

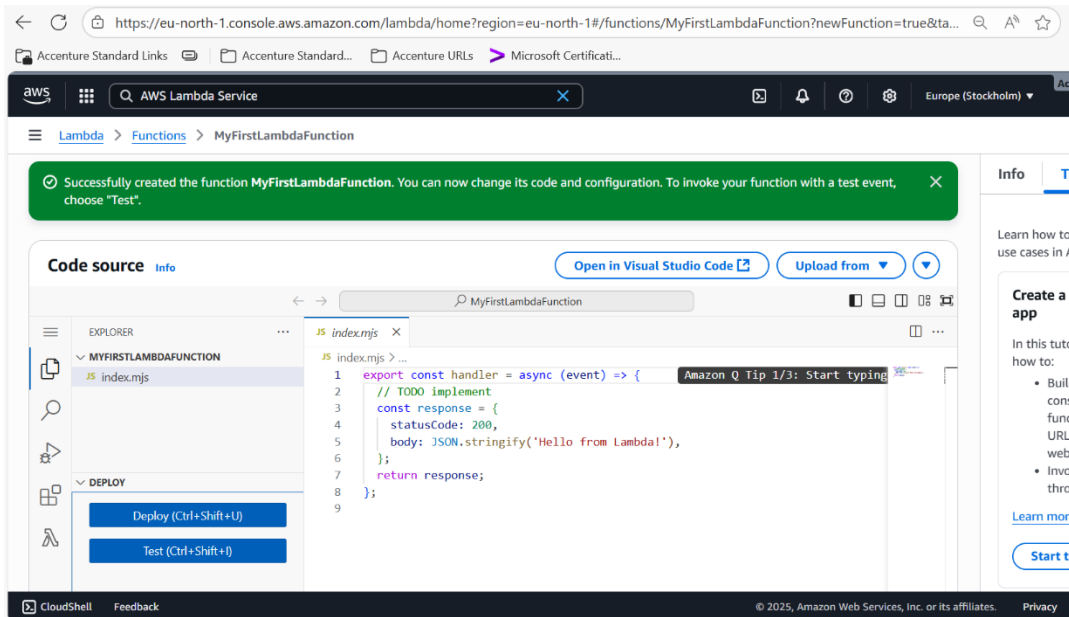
[Learn more](#)

[Start](#)

4

Configure the Runtime and Code

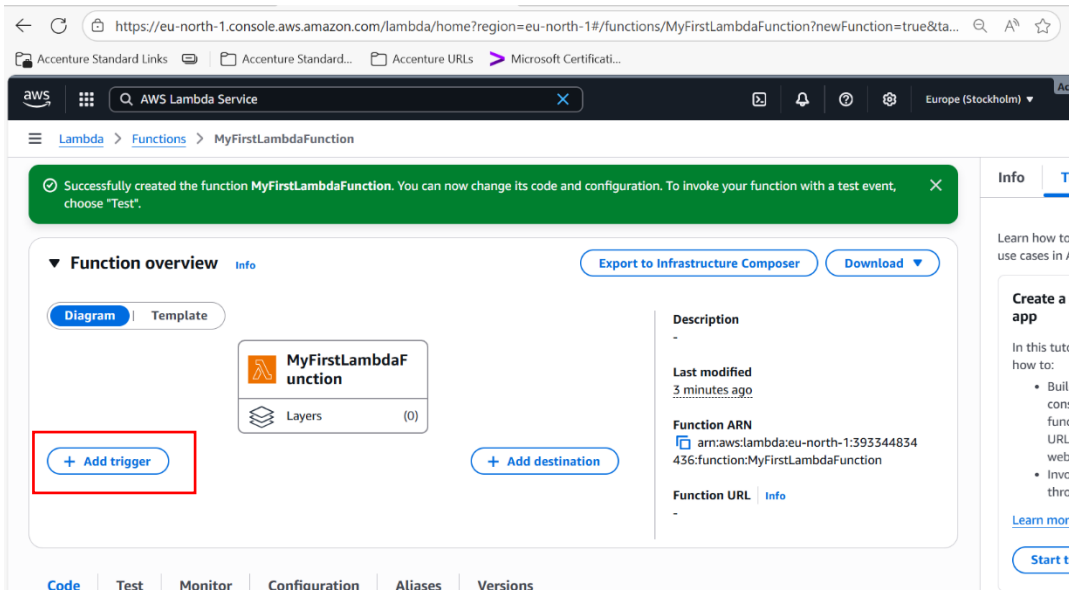
1. After creation, you'll be on the function's configuration page.
2. Use the **Code source** editor to write or paste your function code. The **handler** is the method where execution starts (e.g., `lambda_function.lambda_handler` for Python).
3. **Deployment:** For production, you typically upload a zipped deployment package or use a container image. For a simple exercise, the inline editor is sufficient.
4. **Basic Settings:** Configure memory (MB) and timeout settings under the **Configuration** tab



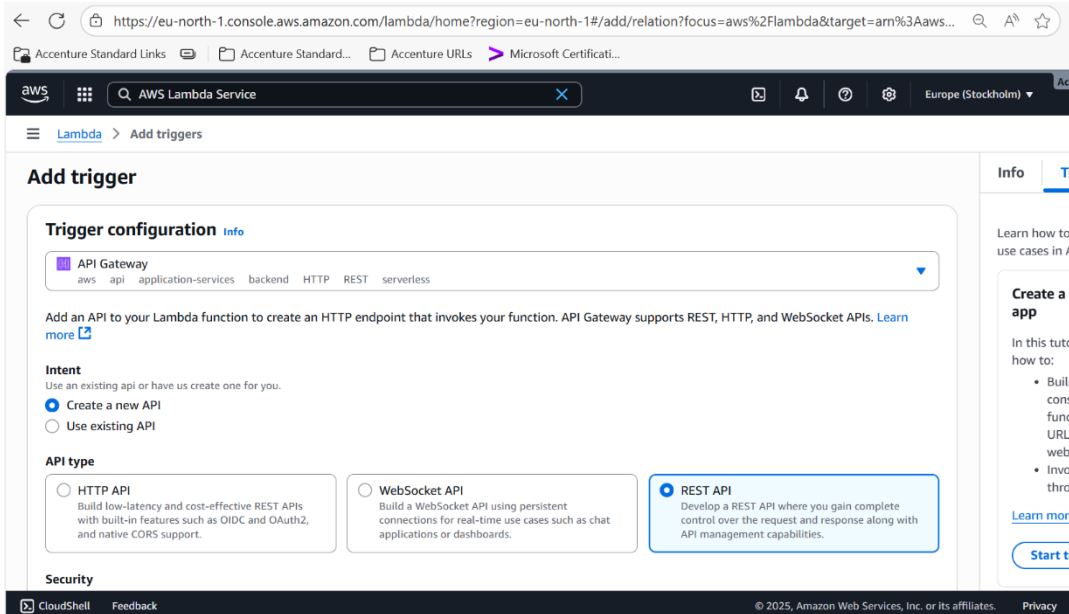
5

Add a Trigger

1. In the **Function overview** on the Lambda console, click **+ Add trigger**.

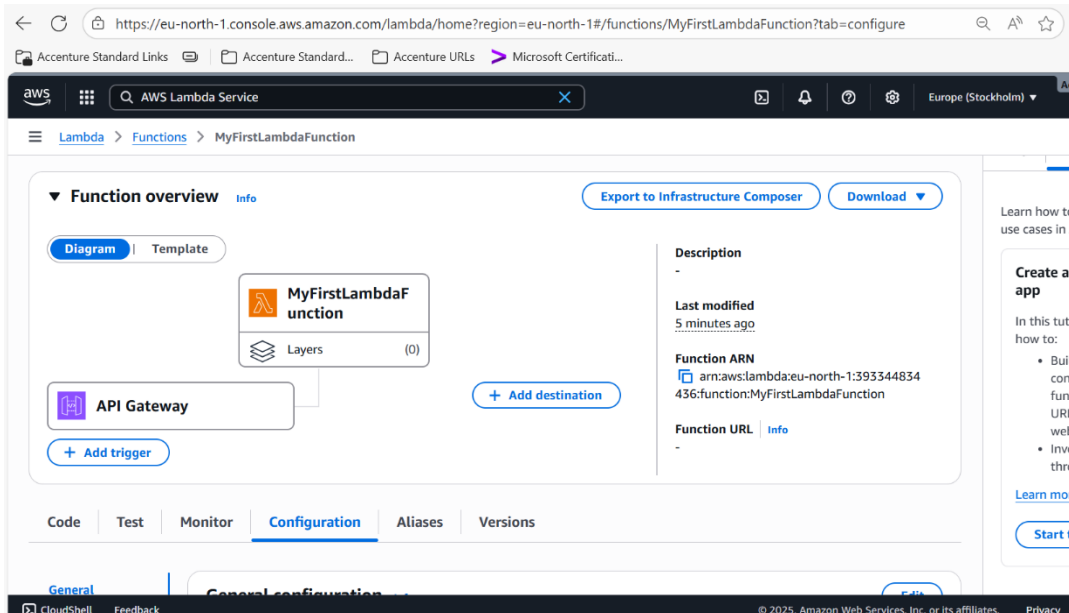


1. Select the service that will trigger the function (e.g., **API Gateway**, **S3**, or **EventBridge**).
2. **Configure the Trigger:** Follow the on-screen prompts, which vary by service. For example:
 - For **S3**, select the target bucket and the event type (e.g., All object create events).
 - For **API Gateway**, choose to **Create an API**, select the API type (e.g., REST API), and security settings.



6

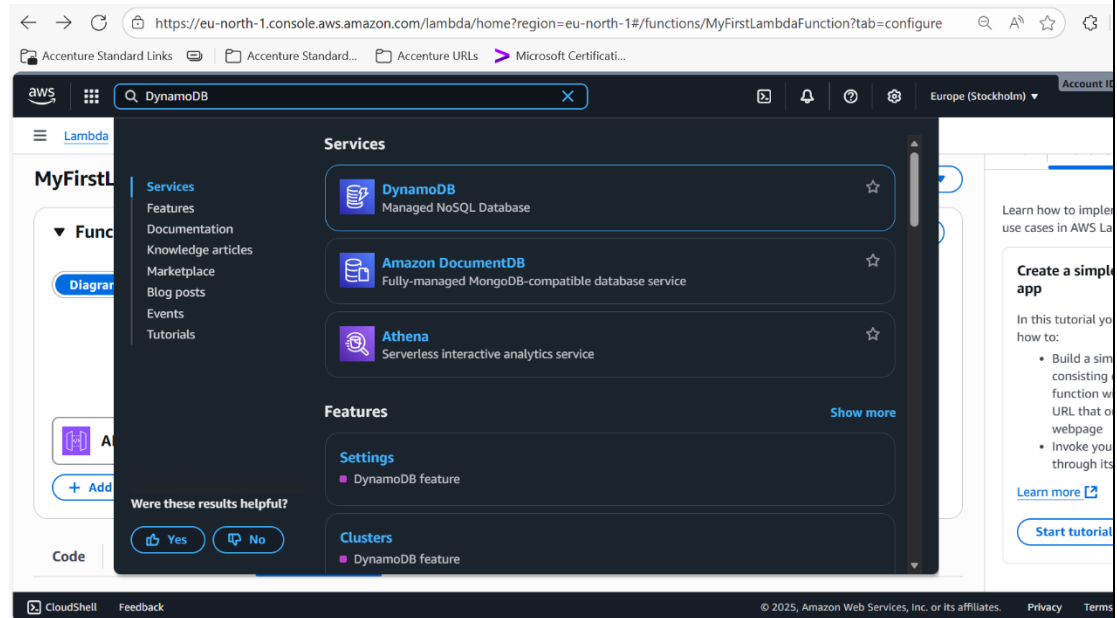
Add



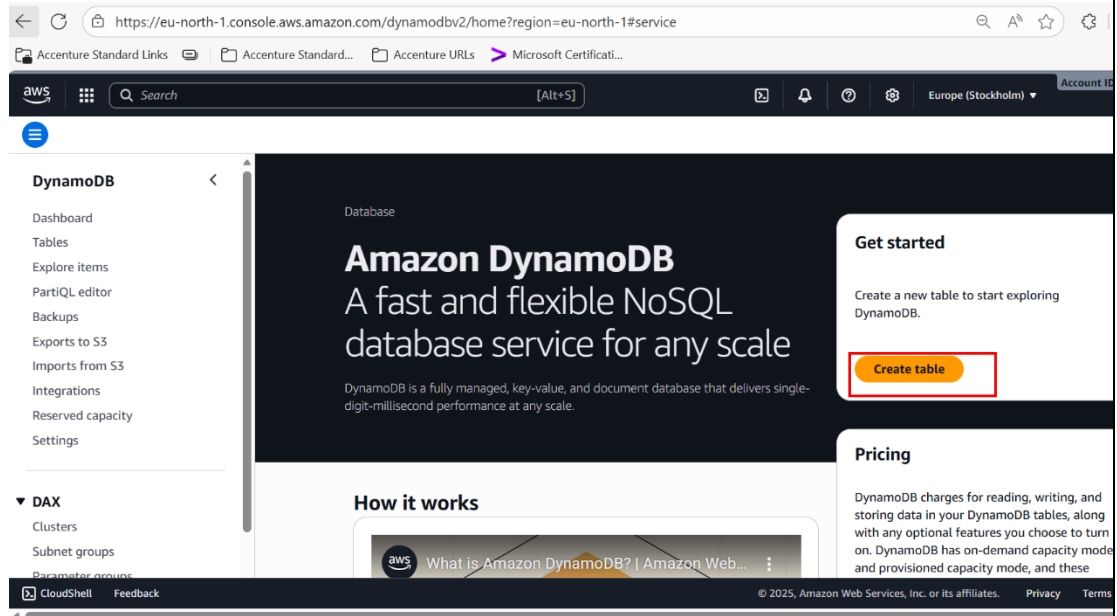
6. Create Resources on AWS

1 Create a NoSQL Database (Amazon DynamoDB)

Navigate to the **DynamoDB** service in the console.



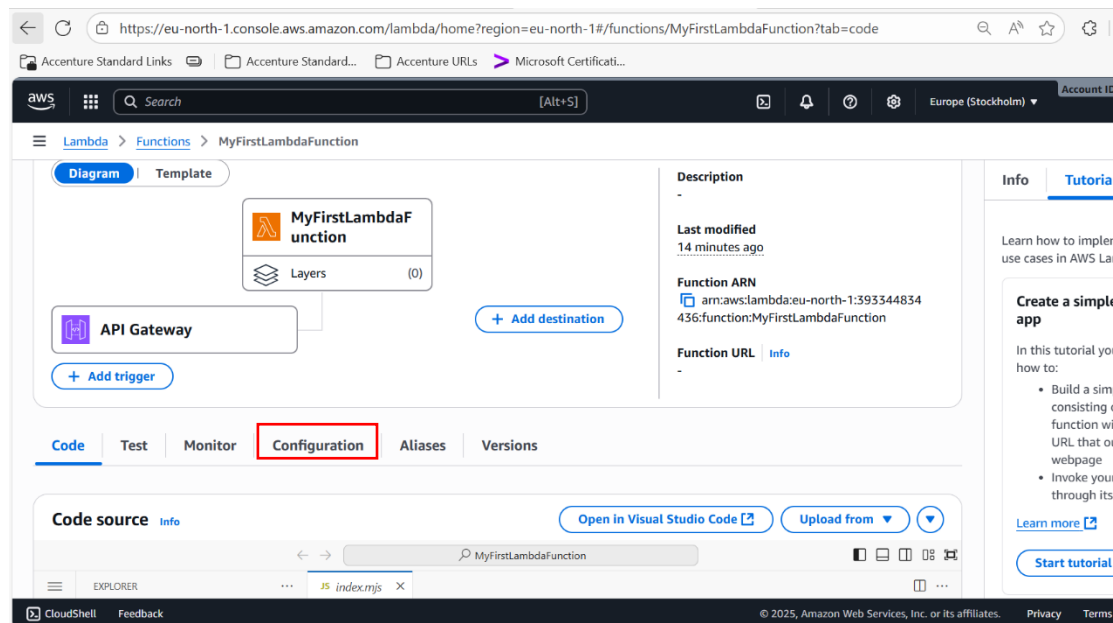
2 Click **Create table**.



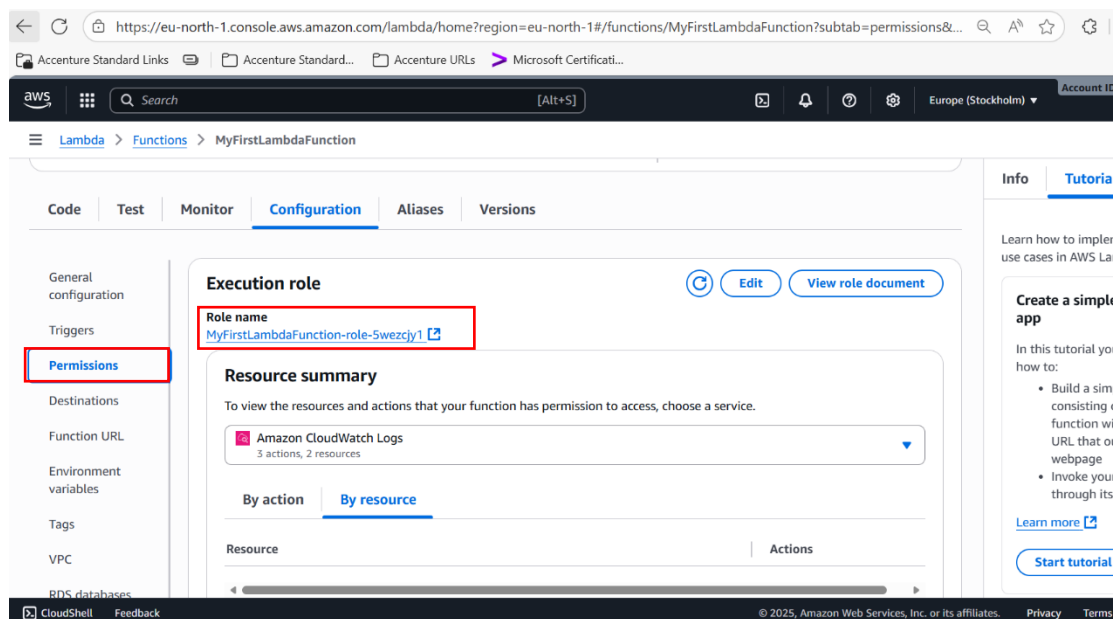
3

Table details:**Table name:** Enter a name (e.g., MyServerlessToDoList).**Partition key:** Define the primary identifier for your items (e.g., UserID, string type).**Sort key (optional):** Define an additional key for more complex lookups (e.g., ItemID, string type).Use default settings for the rest and click **Create table**.**Update Lambda Permissions (IAM Role)**

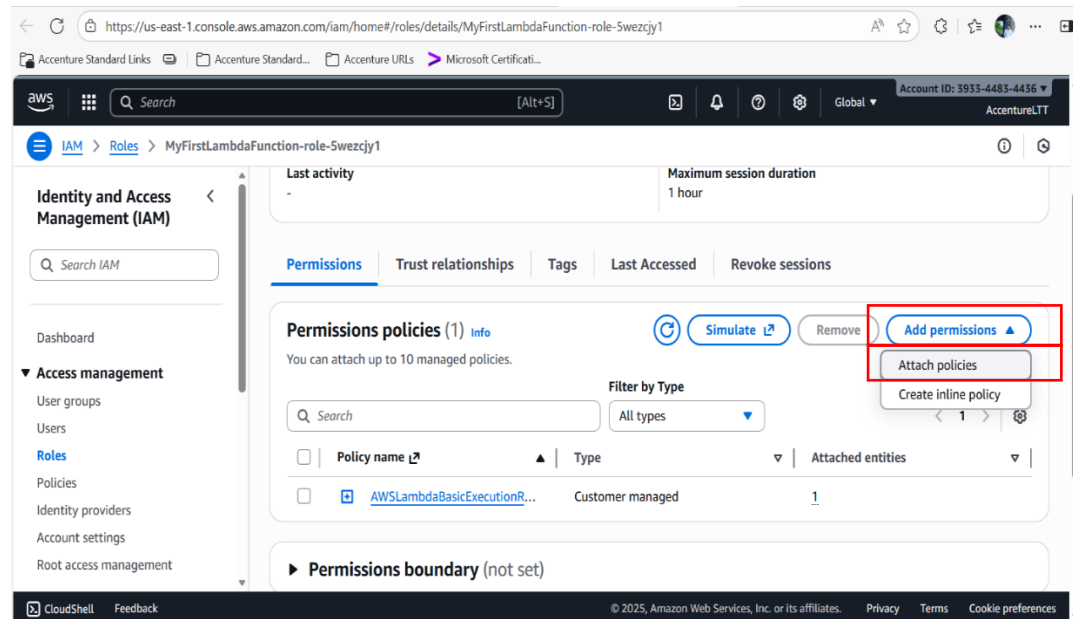
Your Lambda function needs permission to read/write to the new DynamoDB table.

Navigate back to your **Lambda function** in the console.

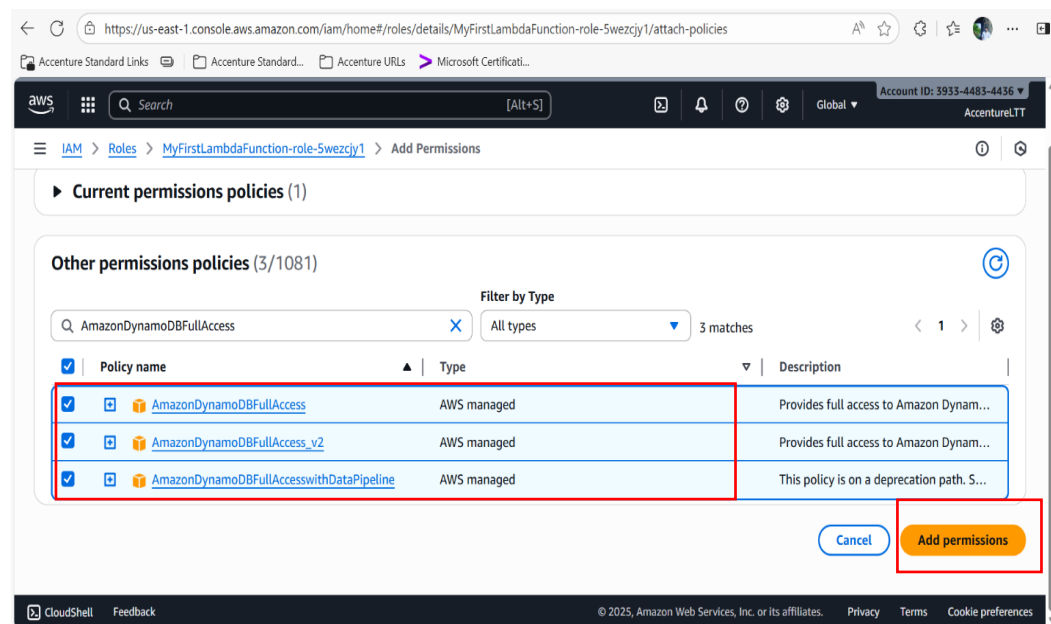
4

Go to the **Configuration** tab and select **Permissions**.

- 5 Click on the **Role name** to open the **IAM Console**.
On the Permissions tab of the role, click **Add permissions** → **Attach policies**.



- 5 Search for a managed policy like **AmazonDynamoDBFullAccess** (or a more restrictive one for production) and attach it.



Integrate and Test

1. **Update Lambda Code:** Modify your Lambda function code to use the AWS SDK (e.g., **Boto3** for Python) to interact with the DynamoDB table. Remember to pass the table name using an **Environment Variable** for best practice.
2. **Add Environment Variable:** In the Lambda configuration, go to **Environment variables** and add a new key-value pair: `TABLE_NAME = ServerlessToDoList`.
3. **Test:** Use the **Test** tab in the Lambda console to invoke your function and verify that an item is added to the DynamoDB table and that the CloudWatch logs show successful execution.

